

Chaos and Analytic Non-Integrability of the Classical Double Pendulum: A Computer-Assisted Proof

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Abstract

The planar double pendulum with two equal point masses on two equal massless rods is the standard example of chaos in classical mechanics, but, as far as we could find, neither its chaos nor its non-integrability has been proved at these parameters: the known proofs need a small parameter (a weak coupling, a small mass ratio, a special link geometry), and the one variational criterion that needs none has been applied only under a parameter condition that, in the text available to us, the equal case does not satisfy. We give a computer-assisted proof. At each of the energies $E = -1/2, 0$ and $1/2$, in units in which the lower rest state has $E = -3$ and $E = 0$ is the energy of releasing both arms from rest in the horizontal position, the flow on the energy level has a hyperbolic periodic orbit with a transversal homoclinic orbit, and the same holds simultaneously for every energy in $[-10^{-10}, 10^{-10}]$. Consequently every real-analytic function on such a level that is invariant under the flow is constant, and the double pendulum has no real-analytic first integral functionally independent of the energy on any connected open set that contains the level $E = 0$. At each of these three energies, explicit covering relations between h-sets give a topological horseshoe (a compact invariant set of the return map that is semiconjugate to a subshift of finite type) with explicit bounds; at $E = 0$ the Poincaré return map has a compact invariant set on which its topological entropy exceeds 0.1016, and the topological entropy of the flow on the level exceeds 0.0138 per unit time. The proofs combine interval arithmetic (the CAPD library), the time-reversal symmetry of the pendulum, cone conditions for the local unstable manifold and covering relations; the programs and their output accompany the paper. Meromorphic non-integrability in the sense of Morales-Ruiz and Ramis remains open.

Keywords: double pendulum; transversal homoclinic orbit; topological horseshoe; topological entropy; non-integrability; computer-assisted proof; interval arithmetic.

MSC 2020: 37J30, 37C29, 37B10, 37N05, 65G20.

1 Introduction

Two equal point masses m hang from two equal massless rods of length l , the first rod from a fixed pivot and the second from the first mass, and swing in a vertical plane under gravity g . With the angles θ_1, θ_2 of the rods from the downward vertical, the Lagrangian is

$$L = ml^2 \left(\dot{\theta}_1^2 + \frac{1}{2} \dot{\theta}_2^2 + \dot{\theta}_1 \dot{\theta}_2 \cos(\theta_1 - \theta_2) \right) + mgl (2 \cos \theta_1 + \cos \theta_2). \quad (1)$$

This is the double pendulum of textbooks and of countless demonstrations, and its chaotic motion is documented numerically and experimentally in detail [22, 24, 26, 13]. A proof has been missing.

The rigorous results we found are of two kinds. The perturbative ones need a small parameter that is not small in the equal case: Burov [4] proves, by the splitting of separatrices, that a physical two-link pendulum has no additional analytic integral when a length in the geometry of the second link is small; Dullin [7] applies Melnikov’s method in the limit of weak coupling, which requires a coupling parameter $\varepsilon \ll 1$, while the mathematical double pendulum with equal masses and lengths corresponds to $\varepsilon = 1/2$ in his normalization (his Sect. 2), and “it can naturally give results close to an integrable case only” (his conclusion, p. 13 of the author’s preprint, which we read; the published text was not available to us); Ivanov conjectures non-integrability for all non-degenerate masses, lengths and energies [9, p. 105], computes homoclinic invariants with a floating-point Runge–Kutta method at three parameter points with mass ratio $m_2/m_1 = 0.01$ [9], and proves transversal homoclinic intersections asymptotically in the limit of a small mass ratio, with the length ratio or the energy near zero or infinity [10, 11], and, for the reduced system obtained in the limit $m_2/m_1 \rightarrow 0$, within explicit distances, between $4 \cdot 10^{-7}$ and $5 \cdot 10^{-3}$, of the four vertices of its compactified parameter square of length ratio and energy [12, Main Theorem, p. 54]; the equal pendulum, with $m_2/m_1 = 1$, is not the reduced system, and none of these statements applies to it; Moauro and Negrini [17] prove transversal homoclinic points and a Bernoulli shift at large energies for a small mass ratio. The one non-perturbative result is Bolotin and Negrini’s variational criterion [2], applied in its Section 10 to the double pendulum near the energy level of the upright equilibrium under an explicit condition on the masses and lengths. In the text of that section available to us, which consists of search snippets only (Sect. 8), the condition from which the theorem is derived fails at equal masses and lengths by a factor of about 2.9, and the authors call it “quite restrictive”. We also searched the computer-assisted-proof literature (arXiv, zbMATH Open, the publication lists of the authors of the CAPD library and its pages of applications; the queries and their results are recorded in the project’s prior-article ledger, available on request): the pendulum results there that we found, by title and abstract, concern the forced damped single pendulum, for example [31], and we found no computer-assisted proof for the double pendulum. On the algebraic side, Stachowiak and Szumiński [27] prove meromorphic non-integrability of restricted double pendula and write that for the classical one “there is no closed mathematical proof confirming its non-integrability” [27, p. 1 of arXiv:1511.01850]; Salnikov [23] computed monodromy matrices numerically at equal masses and lengths; Szumiński and Kapitaniak [28] prove non-integrability of a variable-length double pendulum, a system with three degrees of freedom, and write that “a non-integrability proof for the classical double pendulum is still missing” [28, p. 3 of arXiv:2602.21123]. Kaheman, Bramburger, Kutz and Brunton [13], who build a theory of transport on hypotheses about hyperbolic periodic orbits and their homoclinic connections, note that “the double pendulum lacks such proofs” and that they “lack a proof of transversality of the orbits” [13, Sect. 5.2, p. 27 of arXiv:2209.10132].

We prove the following, at the physical parameters $m_1 = m_2$, $l_1 = l_2$ and without any small parameter. In units $m = l = g = 1$, with the energy H normalized so that the lower rest state has $H = -3$:

- At each of the energies $E = -1/2, 0, 1/2$, the flow on the energy level $M_E = \{H = E\}$ has a hyperbolic periodic orbit, along which the upper arm makes one oscillation and the lower arm one full turn per period, whose stable and unstable manifolds intersect transversally (Theorem 1, computer-assisted). The same holds for every E in $[-10^{-10}, 10^{-10}]$ at once (Theorem 2, computer-assisted).
- Hence every real-analytic function on each of these levels that is invariant under the flow is constant (Corollary 2); and there is no real-analytic first integral on any connected open set containing M_0 that is functionally independent of the energy (Corollary 3).

- At each of the three energies an explicit chain of covering relations gives a topological horseshoe (a compact invariant set semiconjugate to a subshift of finite type; hyperbolicity is not claimed) and quantitative chaos (Theorem 3, computer-assisted, and Corollary 1); at $E = 0$ the return map has a compact invariant set with topological entropy above 0.1016, and the flow on M_0 has topological entropy above 0.0138 per unit time.

The energy $E = 0$ is the energy of the textbook initial condition, both arms released from rest in the horizontal position; it lies in the middle of the band $-1 < E < 1$ in which the lower arm can turn over and the upper arm cannot.

The proof is computer-assisted in the now standard way of rigorous numerics for dynamical systems: every inequality it needs is checked in interval arithmetic with outward rounding, and the flow is enclosed by the rigorous Taylor (Lohner-type) integrators and Poincaré maps of the CAPD library [5]. The part that is written out, rather than computed, uses three ingredients. The time reversal $(\theta, p) \mapsto (-\theta, p)$ of the pendulum reduces the homoclinic problem to finding a single point where the unstable manifold of a symmetric periodic orbit crosses the symmetry line of the section, as in Devaney’s theory of reversible systems [6]. Cone conditions, in the spirit of Zgliczyński’s geometric treatment of the stable manifold theorem [32], control the local unstable manifold as a graph with a certified slope bound; they avoid computing derivatives over several returns, which the interval enclosures of this vector field overestimate strongly. And the covering relations of Zgliczyński and Gidea [33], which have been used for computer-assisted proofs of symbolic dynamics in the restricted three-body problem [30] and in many systems since, give the explicit horseshoe. For the analytic non-integrability we write out Kozlov’s argument [15, Ch. V, §2], which turns an intersection of separatrices that do not coincide into the absence of an analytic integral.

Section 2 sets up the energy levels, the Poincaré section and the reversibility. Section 3 states the results. Section 4 proves Theorems 1 and 2: the lemmas are proved in full, and the computation that verifies their hypotheses is described with its output. Section 5 proves the corollaries, Section 6 the explicit horseshoe, and Section 7 reports negative controls and an independent cross-check. Section 8 discusses what is and is not shown, including the reading of Bolotin and Negrini’s condition. Every statement is labelled: *proved* (a written proof), *computer-assisted* (a written proof whose hypotheses are verified in interval arithmetic by a program that accompanies the paper), or *numerical* (floating-point computation without error control, used only to find the objects and to illustrate).

2 The system, the section and the symmetry

In units $m = l = g = 1$ (time in units of $\sqrt{l/g}$, energy in units of mgl), the Legendre transform of (1) gives, with $c = \cos(\theta_1 - \theta_2)$ and $D = 2 - c^2 = 1 + \sin^2(\theta_1 - \theta_2)$,

$$H(\theta_1, \theta_2, p_1, p_2) = \frac{p_1^2 + 2p_2^2 - 2cp_1p_2}{2D} - 2\cos\theta_1 - \cos\theta_2, \quad (2)$$

on the phase space $T^*\mathbb{T}^2 = \mathbb{T}^2 \times \mathbb{R}^2$. The kinetic energy is $\frac{1}{2}p^\top \mathcal{M}^{-1}p$ with the mass matrix $\mathcal{M} = \begin{pmatrix} 2 & c \\ c & 1 \end{pmatrix}$, which is positive definite. The equations of motion are Hamilton’s equations of (2); the program `check_field.py` derives them from (1) with SymPy and verifies that the vector field given to the rigorous integrator is identical to them (exactly, by reduction of the difference modulo $\cos^2 + \sin^2 = 1$). The lower rest state $\theta = p = 0$ has $H = -3$, and releasing both arms from rest in the horizontal position gives $H = 0$.

Lemma 1 (proved). *Let $-1 < E < 1$. Then E is a regular value of H , and $M_E = H^{-1}(E)$ is a compact connected real-analytic 3-manifold. The set*

$$\Sigma_E = \{x \in M_E : \theta_1 = 0, \dot{\theta}_1 > 0\}$$

is a real-analytic surface in M_E transversal to the flow, and (θ_2, p_2) are coordinates on it: Σ_E is the graph of

$$p_1 = \ell_E(\theta_2, p_2) = \cos \theta_2 p_2 + \sqrt{(1 + \sin^2 \theta_2)(2(E + 2 + \cos \theta_2) - p_2^2)} \quad (3)$$

over the open annulus $A_E = \{(\theta_2, p_2) \in \mathbb{T}^1 \times \mathbb{R} : p_2^2 < 2(E + 2 + \cos \theta_2)\}$.

Proof. A critical point of H has $\partial H / \partial p = \mathcal{M}^{-1}p = 0$, so $p = 0$, and then $(2 \sin \theta_1, \sin \theta_2) = 0$; the four critical points $\theta_1, \theta_2 \in \{0, \pi\}$ have the values $-3, -1, 1, 3$, so every $E \in (-1, 1)$ is regular. $H \geq -3 + \frac{1}{2}\lambda|p|^2$ (the kinetic energy is $\frac{1}{2}p^\top \mathcal{M}^{-1}p$) with $\lambda > 0$ the smallest eigenvalue of $\mathcal{M}^{-1}$ over the compact torus, so H is proper and M_E is compact. The projection of M_E to the configuration torus is $\{V \leq E\}$ with $V = -2 \cos \theta_1 - \cos \theta_2$, and its fibres are ellipses over $\{V < E\}$ and points over $\{V = E\}$, all connected. For fixed θ_2 , $V \leq E$ means $-2 \cos \theta_1 \leq E + \cos \theta_2$; since $-2 < E + \cos \theta_2 < 2$ for $-1 < E < 1$, this is an interval $|\theta_1| \leq \vartheta(\theta_2) < \pi$ of positive length around $\theta_1 = 0$, with ϑ continuous. So $\{V \leq E\}$ is a closed annulus around the circle $\theta_1 = 0$, connected, and M_E , which maps onto it by a closed map with connected fibres, is connected. On $\theta_1 = 0$ we have $c = \cos \theta_2$, $D = 1 + \sin^2 \theta_2$ and $p_1^2 + 2p_2^2 - 2cp_1p_2 = (p_1 - cp_2)^2 + Dp_2^2$, so $H = E$ reads $(p_1 - cp_2)^2 = D(2(E + 2 + \cos \theta_2) - p_2^2)$, while $\dot{\theta}_1 = \partial H / \partial p_1 = (p_1 - cp_2)/D$. The branch $\dot{\theta}_1 > 0$ is (3), defined and analytic exactly on A_E , where $E + 2 + \cos \theta_2 \geq E + 1 > 0$. Transversality is $\dot{\theta}_1 > 0$. $\square$

The return map. Let $P = P_E$ be the first-return map of the flow to Σ_E : $P(z)$ is the first point after z at which the orbit of z crosses $\theta_1 = 0$ upwards. Where the orbit of z reaches this point after crossing $\theta_1 = 0$ only transversally, P and the return time τ are real-analytic near z by the implicit function theorem; the rigorous Poincaré maps of CAPD verify this transversality on every set on which they are evaluated (Sect. 4.4). P preserves the area form $d\theta_2 \wedge dp_2$ (the restriction of $\sum dp_i \wedge d\theta_i$ to $\{\theta_1 = 0\}$); the proofs below do not use this fact. The computation works in the lift $\theta_2 \in \mathbb{R}$, where we write P again for the lifted map, obtained by integrating the flow with θ_2 not reduced modulo 2π , and use

$$\tilde{f} = T \circ P, \quad T(\theta_2, p_2) = (\theta_2 + 2\pi, p_2),$$

which commutes with P because the vector field is 2π -periodic in θ_2 . A fixed point of $\tilde{f}$ is a fixed point of P on the cylinder at which the lower arm makes one full turn backwards per return.

Lemma 2 (reversibility, proved). *Let $G(\theta_2, p_2) = (-\theta_2, p_2)$. If $P(x) = y$, then $P(Gy) = Gx$. Consequently $G \circ P \circ G = P^{-1}$ and $G \circ \tilde{f} \circ G = \tilde{f}^{-1}$ where defined, $\text{Fix}(G) = \{\theta_2 \in \pi\mathbb{Z}\}$ on the cylinder, and if $Gp = p$ is a fixed point of P , then $G(W^u(p)) = W^s(p)$.*

Proof. The involution $R(\theta, p) = (-\theta, p)$ preserves H (c and D are even in the angles) and reverses the symplectic form, so $R \circ \phi_t = \phi_{-t} \circ R$ for the flow ϕ_t . On $\{\theta_1 = 0\}$ it acts as $(0, \theta_2, p_1, p_2) \mapsto (0, -\theta_2, p_1, p_2)$, and ℓ_E is even in θ_2 , so R maps Σ_E to itself and is G in the coordinates (θ_2, p_2) . Let $y = \phi_\tau(x)$ with τ the return time. Then $\phi_s(Ry) = R\phi_{\tau-s}(x)$, whose θ_1 is $-\theta_1(\phi_{\tau-s}x)$ with time derivative $\dot{\theta}_1(\phi_{\tau-s}x)$; an upward crossing of $\theta_1 = 0$ by $\phi_s(Ry)$ at $0 < s < \tau$ would be an upward crossing by the orbit of x at time $\tau - s \in (0, \tau)$, which does not exist. At $s = \tau$ the point

is $Rx \in \Sigma_E$. So $P(Gy) = Gx$, that is $P \circ G \circ P = G$. In the lift the same argument applies to the lifted flow; with $GTG = T^{-1}$ and $TP = PT$ this gives $G\tilde{f}G = T^{-1}P^{-1} = \tilde{f}^{-1}$. If $y \in W^u(p)$, then $P^{-n}(y) \rightarrow p$, and $P^n(Gy) = GP^{-n}(y) \rightarrow Gp = p$, so $Gy \in W^s(p)$; the converse inclusion is the same argument. $\square$

A fixed point with $Gp = p$ is called symmetric. For a fixed point p (of P , or of $\tilde{f}$ in the lift), the unstable set $W^u(p)$ is the set of points z whose backward iterates $P^{-n}(z)$, $n \geq 0$, are all defined and converge to p , and the stable set $W^s(p)$ the set of points z with $P^n(z) \rightarrow p$. The proofs use no global structure of these sets. Near p , Lemma 5 proves that $W^u(p)$ contains a C^1 arc through p , the local unstable manifold $W_{\text{loc}}^u(p)$; when p is symmetric, $W_{\text{loc}}^s(p) = G(W_{\text{loc}}^u(p)) \subset W^s(p)$ by Lemma 2. We say that $W^u(p)$ and $W^s(p)$ intersect transversally at q if there are C^1 arcs through q , one contained in $W^u(p)$ and one in $W^s(p)$, with linearly independent tangents at q ; this is the property that Kozlov's argument in Sect. 5.3 uses.

3 Results

Theorem 1 (computer-assisted). *For each $E \in \{-1/2, 0, 1/2\}$, exactly, the return map P_E has a fixed point $p_E = (0, y_E)$ on $\text{Fix}(G)$, with $\tilde{f}(p_E) = p_E$ in the lift and*

E	y_E	return time T_E	$\text{tr } DP_E(p_E)$
$-1/2$	$[-1.244860970920, -1.244860970917]$	$[3.7047745436, 3.7047745487]$	$[-3.23158, -3.23150]$
0	$[-1.462373092481, -1.462373092479]$	$[2.9534890074, 2.9534890134]$	$[-3.80869, -3.80863]$
$1/2$	$[-1.627044959204, -1.627044959202]$	$[2.5643622220, 2.5643622277]$	$[-3.33283, -3.33277]$

p_E is hyperbolic: $DP_E(p_E)$ has a real eigenvalue of modulus greater than 2.8834, 3.5238 and 2.9987 respectively, and one of modulus less than 1. Its unstable and stable manifolds intersect transversally at a point $q_E \in \text{Fix}(G)$, $q_E \neq p_E$. Equivalently, on the energy level M_E the periodic orbit γ_E through p_E , of period T_E , is hyperbolic and has a transversal homoclinic orbit: its two-dimensional stable and unstable manifolds in M_E intersect transversally along the orbit of q_E .

At $E = 0$, $q_0 = (0 \bmod 2\pi, p_2)$ with $p_2 \in [0.82204913, 0.82225155]$, and $q_0 = P_0^9(z)$ for a point z of the local unstable manifold at distance about $1.9 \cdot 10^{-5}$ from p_0 ; at $E = 1/2$ and $E = -1/2$ the crossing is reached after 11 returns (Table 1).

Theorem 2 (computer-assisted). *For every $E \in [-10^{-10}, 10^{-10}]$ the conclusions of Theorem 1 hold: P_E has a symmetric hyperbolic fixed point $p_E = (0, y_E)$ with*

$$y_E \in [-1.46237309262, -1.46237309234],$$

$$\text{tr } DP_E(p_E) \in [-3.80869, -3.80862],$$

an eigenvalue of modulus greater than 3.5238 and return time in $[2.9534890068, 2.9534890140]$, whose stable and unstable manifolds intersect transversally at a point $q_E \in \text{Fix}(G)$, $q_E \neq p_E$, with $p_2(q_E) \in [0.8220043, 0.8222964]$.

Theorem 3 (computer-assisted). *Let $E \in \{-1/2, 0, 1/2\}$, and let $m = 22$ if $E = 0$ and $m = 26$ if $E = \pm 1/2$. There are $m + 1$ pairwise disjoint compact sets $N, M_0, \dots, M_{m-1}$ in Σ_E (N around p_E , the M_i along the homoclinic loop of q_E) and a compact P_E -invariant set Λ_E contained in their union such that the itinerary map is a continuous surjection from Λ_E onto the subshift of finite type of the graph with the edges $N \rightarrow N$, $N \rightarrow M_0$, $M_i \rightarrow M_{i+1}$ ($0 \leq i \leq m-2$) and $M_{m-1} \rightarrow N$, which semiconjugates $P_E|_{\Lambda_E}$ to the shift (a topological horseshoe: hyperbolicity of Λ_E and a conjugacy are*

not claimed). Moreover $h_{\text{top}}(P_E|_{\Lambda_E}) \geq \log r_m$, with $r_m > 1$ the positive root of $r^{m+1} = r^m + 1$, and the topological entropy of the flow on M_E is at least $\log r_m / \tau_E$, where τ_E bounds the return time on all the sets:

E	r_m	$h_{\text{top}}(P_E _{\Lambda_E}) >$	τ_E	entropy of the flow per unit time $>$
$-1/2$	1.0948327079...	0.0906015	4.2498259	0.0213188
0	1.1069502450...	0.1016087	7.3553854	0.0138141
$1/2$	1.0948327079...	0.0906015	6.9706776	0.0129975

Corollary 1 (proved from Theorem 3). *At each $E \in \{-1/2, 0, 1/2\}$ the return map P_E has a compact invariant set on which it is semiconjugate to a subshift of finite type of positive entropy (a topological horseshoe), and the flow on M_E has positive topological entropy, with the lower bounds of Theorem 3.*

Corollary 2 (proved from Theorems 1 and 2). *For each energy E of Theorems 1 and 2, every real-analytic function on M_E , or on a neighbourhood of M_E in phase space, that is invariant under the flow is constant on M_E .*

Corollary 3 (proved from Theorem 2). *Let $U \subset T^*\mathbb{T}^2$ be a connected open set containing M_0 , and let F be a real-analytic first integral of the double pendulum on U . Then $dF \wedge dH = 0$ on U : F is functionally dependent on H . In particular the double pendulum has no real-analytic first integral on $T^*\mathbb{T}^2$ functionally independent of the energy.*

On the interval of Theorem 2 we prove no horseshoe. The Smale–Birkhoff homoclinic theorem in its later forms, which need no non-resonance condition, would give a horseshoe and positive topological entropy at every energy of that interval from Theorem 2; we have not read those statements and do not use them (Remark 2). The horseshoes of Theorem 3 are proved directly, with explicit bounds, at the three energies.

Figure 1 locates the stored horseshoe design on the section and resolves the image enclosures used to isolate the symmetric crossing.

4 Proof of Theorems 1 and 2

Fix E (or the interval of energies; see Sect. 4.6) and write $f = \tilde{f}$ for the lifted, shifted return map, a real-analytic map on an open subset of $\mathbb{R}^2$. The proof has three steps: a fixed point (Lemma 3), its hyperbolicity and a certified piece of its unstable manifold (Lemmas 4 and 5), and a transversal crossing of that manifold with the symmetry line (Lemma 6). Each lemma is proved here; its hypotheses are inequalities between computable quantities, verified by the programs `prove.cpp` and `cones.cpp` as described in Sect. 4.4.

4.1 The fixed point

Lemma 3 (Krawczyk; proved). *Let $F : U \rightarrow \mathbb{R}^2$ be C^1 on an open set $U \subset \mathbb{R}^2$, $B = p_0 + [-\rho, \rho]^2 \subset U$, C a real 2×2 matrix and $[DF(B)]$ an interval matrix containing $DF(z)$ for all $z \in B$. If*

$$K = p_0 - C F(p_0) + (I - C[DF(B)])[- \rho, \rho]^2,$$

evaluated in interval arithmetic, lies in the interior of B , then F has exactly one zero in B , and it lies in K .

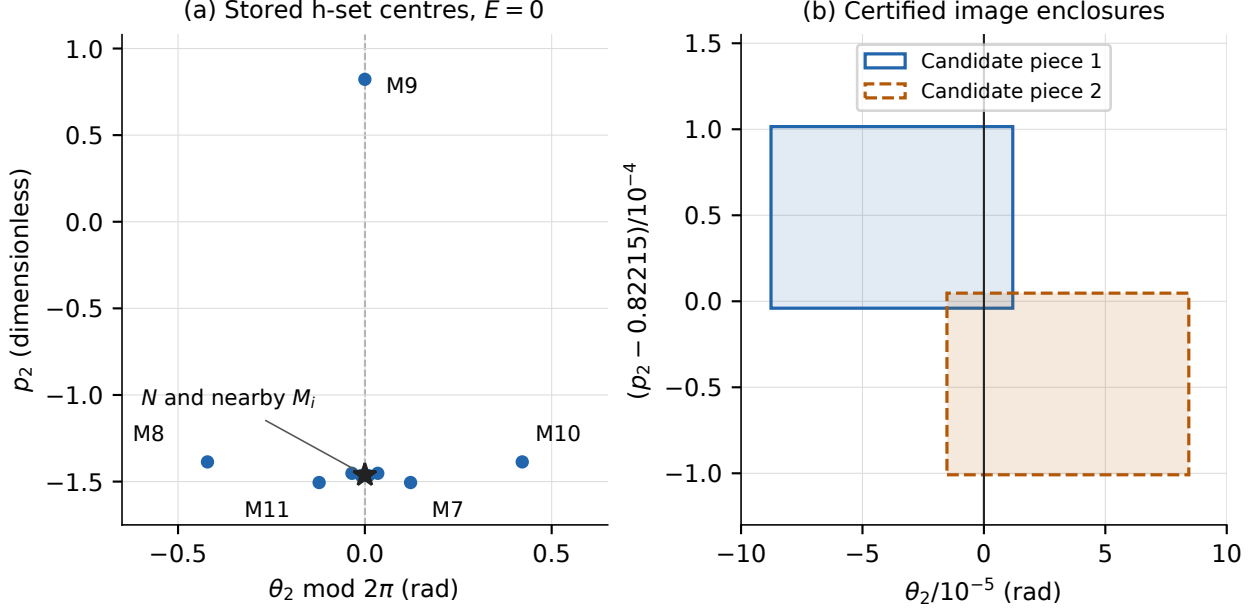


Figure 1: The section geometry at $E = 0$, in the dimensionless units $m = l = g = 1$. (a) Centres of the 23 configured h-sets in `horseshoe_E0.cfg`, with angles reduced modulo 2π ; the star marks the centre of N . These are numerical design points, not a trajectory or a rendering of the invariant manifolds; the small h-set widths are not resolved at this scale. (b) The two rectangular image enclosures printed in `data/E0.txt` for the candidate pieces that may meet the symmetry line $\theta_2 = 0$. Coordinates are rescaled to resolve the certified bounds near q_0 ; overlapping rectangles do not mean that the actual images overlap. The boxes contain images of pieces of the search strip, not just the homoclinic point. Transversality follows from the certified tangent-slope bounds and the lemmas, not from the plotted rectangles. No new integration is used in either panel.

Proof. Write $[M] = I - C[DF(B)]$. The i -th component of $[M][-\rho, \rho]^2$ is $\sum_j [M]_{ij}[-\rho, \rho] = [-s_i, s_i]$ with $s_i = \rho \sum_j \max |[M]_{ij}|$, so $K \subset \text{int } B$ forces $s_i < \rho$, that is $\|I - CJ\|_\infty < 1$ for every matrix J with entries in $[DF(B)]$. Then CJ is invertible for every such J , in particular C is invertible. For $z \in B$ let $J_z = \int_0^1 DF(p_0 + s(z - p_0)) ds$; its entries lie in those of $[DF(B)]$ because intervals are convex. The map $N(z) = z - CF(z)$ satisfies $N(z) = p_0 - CF(p_0) + (I - CJ_z)(z - p_0) \in K \subset B$, so by Brouwer's theorem N has a fixed point $z^* \in K$, and $F(z^*) = 0$ because C is invertible. If $F(z_1) = F(z_2)$ with $z_1, z_2 \in B$, then $0 = J(z_1 - z_2)$ with $J = \int_0^1 DF(z_2 + s(z_1 - z_2)) ds$, whose entries again lie in $[DF(B)]$; CJ is invertible, so $z_1 = z_2$. $\square$

We apply the lemma to $F = f - \text{id}$, with B a box of radius $\rho = 10^{-9}$ around a numerical approximation p_0 on the line $\theta_2 = 0$ and C the floating-point inverse of the midpoint of $[DF(B)]$. The program checks $K \subset \text{int } B$ and, in addition, $G(K) \subset B$. The latter makes the fixed point p symmetric: by Lemma 2, $f(Gp) = Gf^{-1}(p) = Gp$, so Gp is a fixed point of f in B , and uniqueness gives $Gp = p$, that is $p \in \{\theta_2 = 0\}$.

4.2 Cone conditions, hyperbolicity and the local unstable manifold

Fix a real 2×2 matrix $A = [v_u \ v_s]$ whose columns are numerical approximations of the eigenvectors of $Df(p_0)$ ($v_s = Gv_u$ up to sign), and local coordinates $\zeta = (x, y)$, $z = p_0 + A\zeta$, in which we write $\hat{f}(\zeta) = A^{-1}(f(p_0 + A\zeta) - p_0)$ and $\zeta_p = (x_p, y_p)$ for the fixed point. Let $N_0 = [-a, a] \times [-b, b]$ and $\alpha > 0$, and consider the cone $C_u = \{(v_x, v_y) : |v_y| \leq \alpha|v_x|\}$. The conditions are:

- (C1) there are $\mu > 1$ and a sign $\sigma \in \{\pm 1\}$ such that for all $\zeta \in N_0$ and $|t| \leq \alpha$, the vector $(u, w) = D\hat{f}(\zeta)(1, t)$ satisfies $\sigma u \geq \mu$ and $|w| < \alpha|u|$;
- (C2) for all $\zeta \in N_0$: if $|\hat{f}(\zeta)_x| \leq a$, then $|\hat{f}(\zeta)_y| < b$;
- (C3) $|\det Df(z)| < \mu$ for $z \in B$, and $\zeta_p \in \text{int } N_0$.

Lemma 4 (hyperbolicity; proved). *Under (C1) and (C3), $Df(p)$ has a real eigenvalue λ_u with $|\lambda_u| \geq \mu$ and an eigenvector in $\text{int } C_u$, and a second eigenvalue λ_s with $|\lambda_s| < 1$. So p is a hyperbolic fixed point.*

Proof. By (C1) the linear map $M = D\hat{f}(\zeta_p)$ sends every line of slope $t \in [-\alpha, \alpha]$ to a line of slope $w/u \in (-\alpha, \alpha)$; the continuous map $t \mapsto w/u$ of $[-\alpha, \alpha]$ into itself has a fixed point t^* , and $v = (1, t^*)$ is an eigenvector, $Mv = uv$, with $|u| \geq \mu$ and $|t^*| < \alpha$. The other eigenvalue has modulus $|\det M|/|u| < 1$ by (C3); $\det M = \det Df(p)$. (For this area-preserving map $\det Df = 1$ exactly; the lemma does not use it.) $\square$

For the local unstable manifold we need one more condition, a bound on the derivative over the whole box:

- (C4) there is an interval matrix $[M]$ containing $D\hat{f}(\zeta)$ for every $\zeta \in N_0$ such that, with $m_{11} = \min |[M]_{11}|$, $m_{12} = \max |[M]_{12}|$ and $\delta_M = \max |\det [M]|$ (the maximum of $|\det M|$ over the real matrices M with entries in those of $[M]$), the number $\mu_h = m_{11} - \alpha m_{12}$ satisfies

$$\mu_h > 1, \quad \kappa = \frac{\delta_M}{\mu_h} < 1, \quad \vartheta = \frac{\delta_M}{\mu_h^2} < 1, \quad |x_p| < a \frac{\mu_h - 1}{\mu_h + 1}.$$

Here μ_h bounds the expansion of secants in C_u , κ the vertical contraction of the graph transform below, and ϑ the contraction of slopes; the last inequality says that p is central enough in N_0 for graphs through it to be stretched across the box. For our area-preserving map $\det Df = 1$, so δ_M is close to 1 and (C4) says that $|[M]_{11}|$ stays well above 1; the proof does not use area preservation.

Lemma 5 (the graph transform and the local unstable manifold; proved). *Assume that $\hat{f}$ is C^1 on a neighbourhood of N_0 and that (C1)–(C4) hold. Let $\mathcal{L}$ be the set of functions $h : [-a, a] \rightarrow [-b, b]$ with $|h(x) - h(x')| \leq \alpha|x - x'|$, with the distance $\|h_1 - h_2\| = \max |h_1 - h_2|$. For $h \in \mathcal{L}$ write $\text{gr } h = \{(x, h(x)) : |x| \leq a\} \subset N_0$ and $\varphi_h(x) = \hat{f}(x, h(x))_x$.*

- (a) *If $h \in \mathcal{L}$ and $\varphi_h([-a, a]) \supset [-a, a]$, then $|\varphi_h(x) - \varphi_h(x')| \geq \mu|x - x'|$, φ_h is a homeomorphism onto its image, and $\hat{f}(\text{gr } h) \cap ([-a, a] \times \mathbb{R})$ is the graph of a function $\mathcal{H}h \in \mathcal{L}$ with $|\mathcal{H}h| < b$. If h is C^1 , so is $\mathcal{H}h$.*
- (b) *If $\mathcal{H}h_1$ and $\mathcal{H}h_2$ are defined, then $\|\mathcal{H}h_1 - \mathcal{H}h_2\| \leq \kappa\|h_1 - h_2\|$, and $|\varphi_{h_1}^{-1}(\tilde{x}) - \varphi_{h_2}^{-1}(\tilde{x})| \leq c_M\|h_1 - h_2\|$ for $|\tilde{x}| \leq a$, with $c_M = m_{12}/(m_{11} - \alpha m_{12})$.*
- (c) *Let $w \in \mathcal{L}$ with $\mathcal{H}w = w$, and let $h_n \in \mathcal{L}$, $n \geq 0$, be C^1 functions with $h_{n+1} = \mathcal{H}h_n$. Then w is C^1 , and $h_n \rightarrow w$ and $h'_n \rightarrow w'$ uniformly on $[-a, a]$.*
- (d) *There is exactly one $w \in \mathcal{L}$ with $w(x_p) = y_p$ and $\mathcal{H}w = w$. It is C^1 with $|w'| < \alpha$, and $\text{gr } w \subset \hat{f}(\text{gr } w)$. For $\zeta \in \text{gr } w$ and $n \geq 0$ the point $\hat{f}^{-n}(\zeta)$ is defined, lies in $\text{gr } w$ and satisfies $|\hat{f}^{-n}(\zeta)_x - x_p| \leq \mu^{-n}|\zeta_x - x_p|$; so $\hat{f}^{-n}(\zeta) \rightarrow \zeta_p$, and $p_0 + A \text{gr } w \subset W^u(p)$.*

Proof. (The first hypothesis is verified by the computation: stage 2 encloses the first return of every point of N_0 , so $\hat{f}$ is real-analytic on a neighbourhood of N_0 .) The set N_0 is convex. For $\zeta, \zeta' \in N_0$ let $M(\zeta, \zeta') = \int_0^1 D\hat{f}(\zeta' + s(\zeta - \zeta')) ds$; then $\hat{f}(\zeta) - \hat{f}(\zeta') = M(\zeta, \zeta')(\zeta - \zeta')$, and the entries of $M(\zeta, \zeta')$ lie in those of $[M]$ because intervals are convex, so $|M_{11}| \geq m_{11}$, $|M_{12}| \leq m_{12}$ and $|\det M| \leq \delta_M$. If $\zeta - \zeta' = \ell(1, t)$ with $\ell \neq 0$ and $|t| \leq \alpha$, then $M(\zeta, \zeta')(1, t) = \int_0^1 (u_s, w_s) ds$, where $(u_s, w_s) = D\hat{f}(\zeta' + s(\zeta - \zeta'))(1, t)$ satisfies $\sigma u_s \geq \mu$ and $|w_s| < \alpha \sigma u_s$ by (C1). Integrating,

$$\hat{f}(\zeta) - \hat{f}(\zeta') = \ell(\bar{u}, \bar{w}), \quad \sigma \bar{u} \geq \mu, \quad |\bar{w}| < \alpha |\bar{u}| : \quad (4)$$

a secant in the cone C_u is mapped to a secant in C_u , and its x -component is multiplied by a factor of modulus at least μ and sign σ . The same factor $\bar{u} = M_{11} + M_{12}t$ also has modulus at least $m_{11} - \alpha m_{12} = \mu_h$, since $M = M(\zeta, \zeta')$ has its entries in $[M]$.

(a) For $x \neq x'$ in $[-a, a]$ the secant from $(x, h(x))$ to $(x', h(x'))$ lies in C_u , so by (4) $\sigma(\varphi_h(x') - \varphi_h(x))$ has the sign of $x' - x$ and modulus at least $\mu|x' - x|$, and the image secant has slope of modulus less than α . So φ_h is continuous and strictly monotone, and $\hat{f}(\text{gr } h)$ is the graph, over the interval $\varphi_h([-a, a]) \supset [-a, a]$, of a function with Lipschitz constant at most α . Each point of its part over $[-a, a]$ is $\hat{f}(\zeta)$ with $\zeta \in N_0$ and $|\hat{f}(\zeta)_x| \leq a$, so its y -coordinate has modulus less than b by (C2). If h is C^1 , the curve $x \mapsto \hat{f}(x, h(x))$ is C^1 with tangent $D\hat{f}(\xi)(1, h'(x))$, $\xi = (x, h(x))$, whose first component is non-zero by (C1); so φ_h is a C^1 diffeomorphism onto its image and

$$(\mathcal{H}h)'(\varphi_h(x)) = S(\xi, h'(x)), \quad S(\zeta, t) = \frac{w}{u} \text{ for } (u, w) = D\hat{f}(\zeta)(1, t). \quad (5)$$

By (C1), $|S| < \alpha$ on $N_0 \times [-\alpha, \alpha]$, and S is continuous there. Moreover $\partial S / \partial t = \det D\hat{f}(\zeta) / u^2$ with $|u| = |D\hat{f}(\zeta)_{11} + D\hat{f}(\zeta)_{12}t| \geq m_{11} - \alpha m_{12}$, so by the mean value theorem

$$|S(\zeta, t) - S(\zeta, t')| \leq \vartheta |t - t'| \quad (\zeta \in N_0, |t|, |t'| \leq \alpha). \quad (6)$$

(b) Let $|\tilde{x}| \leq a$, $x_i = \varphi_{h_i}^{-1}(\tilde{x})$, $\xi_i = (x_i, h_i(x_i))$, $d = \|h_1 - h_2\|$, and $(\Delta_x, \Delta_y) = \xi_1 - \xi_2$, $M = M(\xi_1, \xi_2)$. Then $M(\Delta_x, \Delta_y) = \hat{f}(\xi_1) - \hat{f}(\xi_2) = (0, \mathcal{H}h_1(\tilde{x}) - \mathcal{H}h_2(\tilde{x}))$. The first row gives $M_{11}\Delta_x = -M_{12}\Delta_y$, so $|\Delta_x| \leq (m_{12}/m_{11})|\Delta_y|$; and $|\Delta_y| \leq |h_1(x_1) - h_2(x_1)| + |h_2(x_1) - h_2(x_2)| \leq d + \alpha|\Delta_x|$. Hence $|\Delta_y| \leq d m_{11} / (m_{11} - \alpha m_{12})$ and $|\Delta_x| \leq c_M d$. Substituting $\Delta_x = -M_{12}\Delta_y / M_{11}$ into the second row gives $\mathcal{H}h_1(\tilde{x}) - \mathcal{H}h_2(\tilde{x}) = (\det M / M_{11})\Delta_y$, of modulus at most $(\delta_M / m_{11}) d m_{11} / (m_{11} - \alpha m_{12}) = \kappa d$.

(c) By (b), $d_n = \|h_n - w\| \leq \kappa^n d_0 \rightarrow 0$. For $|\tilde{x}| \leq a$ let $x^* = \varphi_w^{-1}(\tilde{x})$, $\xi^* = (x^*, w(x^*))$, $x_n = \varphi_{h_n}^{-1}(\tilde{x})$ and $\xi_n = (x_n, h_n(x_n))$. By (b), $|x_n - x^*| \leq c_M d_n$ and $|h_n(x_n) - w(x^*)| \leq d_n + \alpha c_M d_n$, so $\xi_n \rightarrow \xi^*$ uniformly in $\tilde{x}$. On the complete space of continuous functions $s : [-a, a] \rightarrow [-\alpha, \alpha]$ with the maximum distance, the map $(Ts)(\tilde{x}) = S(\xi^*, s(x^*))$ is defined (x^* and ξ^* depend continuously on $\tilde{x}$, and $|S| < \alpha$) and is a contraction with constant ϑ by (6); let s^* be its fixed point. By (5), $h'_{n+1}(\tilde{x}) = S(\xi_n, h'_n(x_n))$, and by (6)

$$|h'_{n+1}(\tilde{x}) - s^*(\tilde{x})| \leq \vartheta |h'_n(x_n) - s^*(x_n)| + \vartheta |s^*(x_n) - s^*(x^*)| + |S(\xi_n, s^*(x^*)) - S(\xi^*, s^*(x^*))|.$$

The last two terms tend to 0 uniformly in $\tilde{x}$, because s^* is uniformly continuous on $[-a, a]$, S is uniformly continuous on the compact set $N_0 \times [-\alpha, \alpha]$, and $x_n \rightarrow x^*$, $\xi_n \rightarrow \xi^*$ uniformly. So $e_n = \|h'_n - s^*\|$ satisfies $e_{n+1} \leq \vartheta e_n + \epsilon_n$ with $\epsilon_n \rightarrow 0$, and $e_n \rightarrow 0$. A uniform limit of C^1 functions whose derivatives converge uniformly is C^1 , with the limit of the derivatives as its derivative: w is C^1 and $h'_n \rightarrow s^* = w'$.

(d) Let $\mathcal{L}_p = \{h \in \mathcal{L} : h(x_p) = y_p\}$. It is closed in the complete space of continuous functions on $[-a, a]$ with the maximum distance, and not empty: $\zeta_p \in \text{int } N_0$ by (C3), so the constant

y_p lies in $\mathcal{L}_p$. For $h \in \mathcal{L}_p$, $\varphi_h(x_p) = x_p$ because $\hat{f}(\zeta_p) = \zeta_p$; by (4), $\varphi_h(a)$ and $\varphi_h(-a)$ lie on opposite sides of x_p at distance at least $\mu_h(a - |x_p|)$, which exceeds $a + |x_p|$ by the last condition in (C4). So $\varphi_h([-a, a]) \supset [-a, a]$, $\mathcal{H}h$ is defined by (a), and $\mathcal{H}h(x_p) = y_p$. Thus $\mathcal{H}$ maps $\mathcal{L}_p$ into itself, is a contraction by (b), and has exactly one fixed point w in $\mathcal{L}_p$. The iterates of the constant y_p under Γ are C^1 by (a), so w is C^1 by (c), and $|w'| < \alpha$ by (5). $\mathcal{H}w = w$ says that $\text{gr } w \subset \hat{f}(\text{gr } w)$: each $\zeta \in \text{gr } w$ is $\hat{f}(\zeta_1)$ with $\zeta_1 = (x_1, w(x_1))$, $x_1 = \varphi_w^{-1}(\zeta_x)$, and $|x_1 - x_p| \leq \mu^{-1}|\zeta_x - x_p|$ by (a) and $\varphi_w(x_p) = x_p$. The map $\hat{f}$ is injective, since P is the first-return map of a flow, so $\zeta_1 = \hat{f}^{-1}(\zeta)$; by induction $\hat{f}^{-n}(\zeta) \in \text{gr } w$ with $|\hat{f}^{-n}(\zeta)_x - x_p| \leq \mu^{-n}|\zeta_x - x_p|$, and $\hat{f}^{-n}(\zeta) \rightarrow \zeta_p$ because $|w(x) - y_p| \leq \alpha|x - x_p|$. $\square$

We call the C^1 arc $W_{\text{loc}}^u(p) = \{p_0 + A(x, w(x)) : |x| \leq a\}$ the local unstable manifold of p ; by (d), $f^{-1}(W_{\text{loc}}^u(p)) \subset W_{\text{loc}}^u(p)$. Lemma 5 is the stable manifold theorem (for diffeomorphisms of a manifold see, e.g., [8, Sect. 4.1, Thm. 4]) for this map, proved by the graph transform with constants certified by (C1)–(C4), so that no statement for maps of a whole manifold has to be applied to a map defined only near p . Zgliczyński [32] treats the stable manifold theorem with covering relations and cone conditions; we do not use his statements.

4.3 The symmetric transversal crossing

Let $[x_1, x_2] \subset [-a, a]$ with $x_p \notin [x_1, x_2]$, let $\bar{\rho} \geq \max\{|x - x_p| : x \in [x_1, x_2]\}$, and let $\mathcal{R} = [x_1, x_2] \times Y_{\mathcal{R}}$, $Y_{\mathcal{R}} = [y_p - \alpha\bar{\rho}, y_p + \alpha\bar{\rho}]$, which contains the graph of w over $[x_1, x_2]$ by Lemma 5 and the mean value theorem. For $k \geq 1$ and an integer m consider:

- (C5) the θ_2 -components of f^k on the two edges $\{x_1\} \times Y_{\mathcal{R}}$ and $\{x_2\} \times Y_{\mathcal{R}}$ of $\mathcal{R}$ (mapped by $\zeta \mapsto p_0 + A\zeta$) lie strictly on opposite sides of $m\pi$;
- (C6) $\mathcal{R}$ is covered by pieces Z such that, for every piece whose image $f^k(Z)$ may meet the line $\theta_2 = m\pi$, every vector $Df^k(p_0 + A\zeta)A(1, t)$ with $\zeta \in Z$ and $|t| \leq \alpha$ has both components non-zero.

Lemma 6 (symmetric transversal homoclinic point; proved). *Under the hypotheses of Lemma 5 and (C5)–(C6), $W^u(p)$ and $W^s(p)$ (for the return map on the cylinder) intersect transversally at a point $q \in \text{Fix}(G)$, $q \neq p$.*

Proof. The arc $\Gamma = \{p_0 + A(x, w(x)) : x_1 \leq x \leq x_2\} \subset W^u(p)$ runs from one edge of $\mathcal{R}$ to the other, and $f^k(\Gamma)$ is a continuous arc in the lift whose θ_2 changes sides of $m\pi$ by (C5). So there is $x^* \in (x_1, x_2)$ with $q = f^k(z)$, $z = p_0 + A(x^*, w(x^*))$, on the line $\theta_2 = m\pi$; on the cylinder $q \in \text{Fix}(G)$. q is in $W^u(p)$, and $q = Gq \in G(W^u(p)) = W^s(p)$ by Lemma 2. $q \neq p$ on the cylinder: P^k is injective, $P^k(p) = p$, and $z \neq p$ because $x^* \neq x_p$ and both points lie in a box of size far below 2π . The tangent of $W^u(p)$ at q is $v = Df^k(z)A(1, w'(x^*))$, and $|w'(x^*)| \leq \alpha$; the piece of (C6) that contains $(x^*, w(x^*))$ has an image that meets the line, so both components of $v = (v_1, v_2)$ are non-zero. So $f^k(\Gamma) \subset W^u(p)$ is a C^1 arc through q with tangent v (f^k is a diffeomorphism near Γ , and Γ is C^1 by Lemma 5), and $G(f^k(\Gamma)) \subset W^s(p)$ (Lemma 2) is a C^1 arc through $Gq = q$ with tangent $DGv = (-v_1, v_2)$. Then $\det[v, DGv] = 2v_1v_2 \neq 0$: the intersection is transversal in the sense of Sect. 2. For the flow, $W^u(\gamma)$ and $W^s(\gamma)$ are the flow-saturations of $W^u(p)$ and $W^s(p)$, so their tangent planes at q are spanned by the vector field and the two section tangents, and they intersect transversally in M_E along the orbit of q . $\square$

4.4 The computation

The program `prove.cpp` (with the header `rig.h`) verifies $K \subset \text{int } B$ and $G(K) \subset B$ (stage 1), (C1)–(C3) (stage 2) and (C5)–(C6) (stage 3), in interval arithmetic in double precision with outward rounding, using CAPD [5] at a fixed commit (03dc5628 of <https://github.com/CAPDGroup/CAPD>). It reports **FAIL** and exits with a non-zero status if any check fails.

Enclosures of the return map. A point or a small box of (θ_2, p_2) is lifted to phase space by (3) and integrated by CAPD’s rigorous Taylor method of order 20 with a Lohner-type representation of sets. CAPD’s Poincaré map for the section $\theta_1 = 0$ with crossing direction “minus to plus” returns an enclosure of the first upward crossing and of the return time; its C^1 version also returns an enclosure of the derivative of the four-dimensional Poincaré map, including the correction for the variation of the return time. The derivative of f in (θ_2, p_2) is $\Pi DP_4 L$, with Π the projection onto (θ_2, p_2) and L the Jacobian of the lift, $\partial \ell_E / \partial (\theta_2, p_2)$ enclosed over the set. Every set passed to the C^1 computation contains the centre point used by the mean-value form, so that $f(\zeta) \in f(\zeta_c) + [Df(Z)](\zeta - \zeta_c)$ is valid on every piece Z . A set whose first return cannot be enclosed (for example because the flow is not provably transversal to the section on a step enclosure on which the section function takes the value 0) makes the program fail on that set.

Stage 2. N_0 is covered by a grid of sub-boxes that overlap by a relative 10^{-6} . On each sub-box the program encloses $D\hat{f} = A^{-1}[Df]A$ and checks (C1) for the whole interval $t \in [-\alpha, \alpha]$ (with the sign of u the same on every sub-box), and (C2) by the mean-value enclosure of $\hat{f}$ on the sub-box. (C3) is checked with the enclosure of ζ_p from stage 1. Stage 2 also checks $|y_p| + \alpha(a + |x_p|) < b$, which the proofs do not use.

Stage 3. $[x_1, x_2]$ is covered by pieces, bisected where the check does not resolve. For a piece Z , the images $f^i(Z)$, $1 \leq i \leq k$, are enclosed by carrying one Lohner set through k successive returns, $Y_i \ni f^i(Z)$. The tangent directions in (C6) are propagated through the chain rule $Df^k(z) = Df(f^{k-1}z) \cdots Df(fz) Df(z)$ with one-return derivative enclosures over $Z, Y_1, \dots, Y_{k-1}$, never with a derivative enclosure over several returns: a set of lines $\{(1, s) : s \in S\}$ is mapped by an interval matrix to the hull of the interval quotients of its image over 16 sub-intervals of S , a valid enclosure as long as the first components stay away from 0 (otherwise the inverse slope is used). This choice is forced by a diagnostic measurement, not part of the proof and not recorded in `data/`: over one return the width of CAPD’s derivative enclosure is about $5 \cdot 10^4$ times the width of the initial box, and products of interval matrices over several returns overestimate further; the local box N_0 is therefore thin ($a = 2.5 \cdot 10^{-5}$, $b = 4 \cdot 10^{-8}$ at $E = 0$). The program asserts that the chain has exactly k factors.

Condition (C4). The program `cones.cpp` covers N_0 by 100 sub-boxes of full height that overlap by a relative 10^{-6} , encloses $A^{-1}[Df]A$ on each with the same one-return C^1 computation, and takes for $[M]$ the interval hull of these enclosures; m_{11} , m_{12} , δ_M , μ_h and the four inequalities of (C4) are evaluated in interval arithmetic, with the enclosure of ζ_p from a repetition of stage 1 in which, unlike `prove.cpp` (which compares K with the outward rounding of B), K and $G(K)$ are compared with inward-rounded bounds of B , as Lemma 3 requires. It takes about ten seconds per energy on two threads; the reports are `data/cones_*.txt`. At the three energies and on the interval, $|x_p| \leq 1.99 \cdot 10^{-12}$, $1.69 \cdot 10^{-12}$, $1.87 \cdot 10^{-12}$ and $5.36 \cdot 10^{-10}$, against $a(\mu_h - 1)/(\mu_h + 1) \geq 8.71 \cdot 10^{-6}$, $1.39 \cdot 10^{-5}$, $6.49 \cdot 10^{-6}$ and $1.39 \cdot 10^{-5}$.

Output. Table 1 lists the configuration and the verified quantities; the full reports are `data/E0.txt`, `data/Ehalf.txt` and `data/Eminushalf.txt`. At every energy the local unstable manifold is followed on the branch through $-v_u$, and the crossing is with the line $\theta_2 = 0$ (in the lift, after the shift, $m = 0$). The crossing points differ in kind: $p_2 \approx 0.82$ at $E = 0$ and $1/2$, and $p_2 \approx -1.466$ at $E = -1/2$, where the proof uses another symmetric homoclinic point (Sect. 8).

	$E = -1/2$	$E = 0$	$E = 1/2$
N_0 : a , b ; grid	$1.8 \cdot 10^{-5}$, $3 \cdot 10^{-8}$; 800	$2.5 \cdot 10^{-5}$, $4 \cdot 10^{-8}$; 1000	$1.3 \cdot 10^{-5}$, $2 \cdot 10^{-8}$; 600
cone α ; μ	10^{-3} ; 2.88344	10^{-3} ; 3.52384	10^{-3} ; 2.99874
largest $ w / u $ on N_0	$6.04 \cdot 10^{-4}$	$3.65 \cdot 10^{-4}$	$3.04 \cdot 10^{-4}$
(C4): κ ; ϑ	0.3547; 0.1233	0.2878; 0.0818	0.3351; 0.1118
$[x_1, x_2]$	$[-1.4256, -1.4240] \cdot 10^{-5}$	$[-1.8870, -1.8838] \cdot 10^{-5}$	$[-1.0330, -1.0312] \cdot 10^{-5}$
k	11	9	11
θ_2 of f^k (edge x_1)	$[-8.050, -7.878] \cdot 10^{-4}$	$[1.519, 1.522] \cdot 10^{-3}$	$[3.693, 3.707] \cdot 10^{-3}$
θ_2 of f^k (edge x_2)	$[8.915, 9.087] \cdot 10^{-4}$	$[-3.119, -3.115] \cdot 10^{-3}$	$[-5.347, -5.333] \cdot 10^{-3}$
pieces checked; meeting the line	80; 9	34; 2	66; 1
slope $dp_2/d\theta_2$ there	$[-8.2916, -0.0936]$	$[-6.2138, -0.5237]$	$[-21.416, -0.2789]$
p_2 at the crossing	$[-1.4657285, -1.4656772]$	$[0.8220491, 0.8222516]$	$[0.8244028, 0.8245038]$

Table 1: Theorem 1: configuration and verified quantities (computer-assisted; the numbers are rounded outward from `data/*.txt`; μ is the six-digit value printed by `prove.cpp`, which rounds to nearest, lowered by one unit in its last digit, hence a lower bound for the μ verified in (C1); the largest $|w|/|u|$ is a floating-point figure reported for information, not a bound; κ and ϑ are rounded up).

4.5 Proof of Theorem 1

At each energy, stage 1 gives the fixed point p in the enclosure listed in the theorem, on $\text{Fix}(G)$, with the trace and return time enclosed over B ; Lemma 4 gives hyperbolicity with $|\lambda_u| \geq \mu$; Lemmas 5 and 6 give the transversal homoclinic point $q_E \in \text{Fix}(G)$. The fixed point of $\tilde{f}$ is a fixed point of P on the cylinder, and in one return θ_2 decreases by exactly 2π (the lower arm turns once) while θ_1 makes one oscillation, from 0 back to 0, without reaching $\pm\pi$ (the upper arm cannot turn over: $V(\pm\pi, \theta_2) = 2 - \cos \theta_2 \geq 1 > E$). $\square$

4.6 An interval of energies: proof of Theorem 2

The energy enters the computation only through the lift (3). The configuration `E0_interval.cfg` is that of $E = 0$ with E replaced by the interval $[-10^{-10}, 10^{-10}]$ (its endpoints are enclosed as $\pm 1/10^{10}$ in interval arithmetic). Every enclosure is then valid simultaneously for all E in the interval, and so is every verified inequality: the Krawczyk operator encloses K_E for each E , so each F_E has a unique zero in B , symmetric by the same argument, and stages 2 and 3 verify (C1)–(C3), (C5) and (C6) for each E ; `cones.cpp` verifies (C4) for each E in the same way. Lemmas 3–6 applied at each E give the theorem. The report is `data/E0_interval.txt`: stage 1 gives K with θ_2 -component in $[-5.81, 5.81] \cdot 10^{-10}$ and p_2 -component in $[-1.46237309262, -1.46237309234]$, inside B ($\rho = 10^{-9}$), and $G(K) \subset B$; stage 2 gives $\mu \geq 3.52383$ (the printed 3.52384 lowered by one unit) and the largest cone ratio about $3.66 \cdot 10^{-4}$ on the same N_0 as at $E = 0$; stage 3, on the same segment with $k = 9$, gives edge images with $\theta_2 \in [1.5069, 1.5341] \cdot 10^{-3}$ and $[-3.1300, -3.1031] \cdot 10^{-3}$, and on the 14 of 74 pieces whose image may meet the line (after bisection to depth 4) the slope $dp_2/d\theta_2 \in [-20.744, -0.4104]$; and `data/cones_E0_interval.txt` gives $\kappa \leq 0.2879$ and $\vartheta \leq 0.0818$. The same computation with the interval $[-10^{-9}, 10^{-9}]$ fails stage 1 with the same box B (the θ_2 -component of K is then $[-5.79, 5.79] \cdot 10^{-9}$, ten times wider; `data/stage1_interval_1e-9.txt`, from the repetition of stage 1 in `cones.cpp`); a wider interval of energies needs a larger box or a cover of the interval by several runs. $\square$

5 Proofs of the corollaries

5.1 Entropy of the flow from a section

Topological entropy is taken in Bowen's form. For a continuous map g of a compact metric space (Y, d) , a set $E \subset Y$ is (n, ϵ) -separated if any two distinct $x, y \in E$ have $d(g^j x, g^j y) \geq \epsilon$ for some $0 \leq j < n$; with $s_n(g, \epsilon)$ the largest cardinality of such a set, $h_{\text{top}}(g) = \lim_{\epsilon \rightarrow 0} \limsup_n \frac{1}{n} \log s_n(g, \epsilon)$ [29, Sect. 7.2]. For a continuous flow ϕ on a compact metric space, E is (T, δ) -separated if any two distinct $x, y \in E$ have $d(\phi_s x, \phi_s y) \geq \delta$ for some $s \in [0, T]$; with $M_T(\phi, \delta)$ the largest cardinality, $h(\phi) = \lim_{\delta \rightarrow 0} \limsup_{T \rightarrow \infty} \frac{1}{T} \log M_T(\phi, \delta)$ [3, p. 185]. (Strict or non-strict inequalities give the same limits.) On M_E we use any metric compatible with its topology, for instance the one induced by $(\theta, p) \mapsto (\cos \theta_1, \sin \theta_1, \cos \theta_2, \sin \theta_2, p_1, p_2) \in \mathbb{R}^6$, and on $\Lambda \subset \Sigma_E \subset M_E$ its restriction; topological entropy on a compact space does not depend on this choice.

Lemma 7 (flow entropy from a section; proved). *Let $\Lambda \subset \Sigma_E$ be compact with $P(\Lambda) = \Lambda$, P continuous and injective on Λ and the return time τ continuous on Λ , and let $\tau_{\max} = \max_{\Lambda} \tau$ and $K = \{\phi_s(z) : z \in \Lambda, 0 \leq s \leq \tau(z)\}$. Then K is compact and invariant under the flow, and*

$$h_{\text{top}}(\phi|_K) \geq h(\phi|_K) \geq h_{\text{top}}(P|_{\Lambda})/\tau_{\max}.$$

In particular $h_{\text{top}}(\phi_1) \geq h_{\text{top}}(P|_{\Lambda})/\tau_{\max}$ for the time-one map of the flow on M_E .

Proof. K is the image of the compact set $\{(z, s) : z \in \Lambda, 0 \leq s \leq \tau(z)\}$ under $(z, s) \mapsto \phi_s(z)$, hence compact. Since $\phi_{\tau(z)}(z) = P(z) \in \Lambda$ and every point of Λ is P of a point of Λ , the orbit of each point of K runs through the segments $\{\phi_s(P^j z)\}$, $j \in \mathbb{Z}$, and stays in K . For $z \in \Lambda$ let $t_j(z) = \sum_{0 \leq i < j} \tau(P^i z)$ be its j -th return time; its forward orbit meets Σ_E exactly at these times, because P is the first-return map. τ is continuous and positive on the compact set Λ , so $\tau_{\min} = \min_{\Lambda} \tau > 0$ (there is no equilibrium on M_E : the critical values of H are $-3, -1, 1, 3$, Lemma 1). Fix $0 < \eta < \tau_{\min}/2$.

(a) The map $\Psi(z, \rho) = \phi_{\rho}(z)$ on $\Lambda \times [-\eta, \eta]$ is injective: if $\phi_{\rho}(z) = \phi_{\rho'}(z')$ with $\rho > \rho'$, then $z' = \phi_{\rho - \rho'}(z) \in \Sigma_E$ with $0 < \rho - \rho' < \tau_{\min} \leq \tau(z)$, an earlier return; so $\rho = \rho'$ and $z = z'$. Its image lies in K (for $\rho < 0$ write $z = P(z_{-1})$, $z_{-1} \in \Lambda$, and $\phi_{\rho}(z) = \phi_{\tau(z_{-1}) + \rho}(z_{-1})$). Being a continuous injection of a compact space, Ψ is a homeomorphism onto its image K_{η} , and Ψ^{-1} is uniformly continuous: for every $\epsilon > 0$ there is $\delta_1(\epsilon) > 0$ such that $d(\Psi(z, \rho), \Psi(z', \rho')) < \delta_1(\epsilon)$ implies $d(z, z') < \epsilon$.

(b) The set $K' = \{\phi_s(z) : z \in \Lambda, \eta \leq s \leq \tau(z) - \eta\}$ is compact and disjoint from Λ (a point of $\Lambda \subset \Sigma_E$ equal to $\phi_s(z)$ with $0 < s < \tau(z)$ would be an earlier return), so $\delta_2 = \text{dist}(\Lambda, K') > 0$. Every point of K lies in $K_{\eta} \cup K'$ (for $\tau(z) - \eta < s \leq \tau(z)$, $\phi_s(z) = \phi_{s - \tau(z)}(Pz)$). Hence a point $k \in K$ with $\text{dist}(k, \Lambda) < \delta_2$ is $\Psi(z, \rho)$ with $z \in \Lambda$ and $|\rho| < \eta$ (if $|\rho| = \eta$ then $k \in K'$ as well).

(c) Let $\epsilon > 0$, $\delta = \min(\delta_1(\epsilon), \delta_2)$, and $x, y \in \Lambda$ with $d(\phi_s x, \phi_s y) < \delta$ for all $s \in [0, T]$. We show by induction on j that for every j with $t_j(x) \leq T$: $|t_j(y) - t_j(x)| < \eta$ and $d(P^j x, P^j y) < \epsilon$. For $j = 0$ this holds since $d(x, y) < \delta \leq \delta_1(\epsilon) \leq \epsilon$ (we may take $\delta_1(\epsilon) \leq \epsilon$). Suppose it holds for j , and let $u = t_{j+1}(x) \leq T$. Since $\phi_u(x) = P^{j+1}x \in \Lambda$ and $d(\phi_u y, \phi_u x) < \delta_2$, (b) gives $\phi_u(y) = \Psi(z'', \sigma)$ with $z'' \in \Lambda$, $|\sigma| < \eta$; then $z'' = \phi_{u - \sigma}(y) \in \Sigma_E$ is a return of y , $z'' = P^i y$ with $t_i(y) = u - \sigma$. First, $i > j$: $t_i(y) > t_j(x) + \tau_{\min} - \eta > t_j(x) + \eta > t_j(y)$. Second, $i = j + 1$: otherwise the return $s^* = t_{j+1}(y)$ satisfies $t_j(y) + \tau_{\min} \leq s^* \leq t_i(y) - \tau_{\min} < u \leq T$, so $d(\phi_{s^*} x, \phi_{s^*} y) < \delta_2$ with $\phi_{s^*}(y) = P^{j+1}y \in \Lambda$, and by (b) $\phi_{s^*}(x) = \Psi(z''', \rho)$, $|\rho| < \eta$, $z''' \in \Lambda$: a return of x at the time $s^* - \rho$, which lies strictly between $t_j(x)$ and $t_{j+1}(x)$ (it is $> t_j(x) - \eta + \tau_{\min} - \eta > t_j(x)$ and $< u - \sigma - \tau_{\min} + \eta < u$, since $2\eta < \tau_{\min}$), impossible for consecutive returns. So $t_{j+1}(y) = u - \sigma$, and $d(\Psi(P^{j+1}x, 0), \Psi(P^{j+1}y, \sigma)) = d(\phi_u x, \phi_u y) < \delta_1(\epsilon)$ gives $d(P^{j+1}x, P^{j+1}y) < \epsilon$.

(d) Since $t_j(x) \leq j\tau_{\max}$, (c) with $T = (n-1)\tau_{\max}$ shows: if $x, y \in \Lambda$ have $d(P^j x, P^j y) \geq \epsilon$ for some $j < n$, then $d(\phi_s x, \phi_s y) \geq \delta$ for some $s \in [0, (n-1)\tau_{\max}]$. So an (n, ϵ) -separated set of $P|_\Lambda$ is $((n-1)\tau_{\max}, \delta)$ -separated for the flow on K , and $M_{(n-1)\tau_{\max}}(\phi|_K, \delta) \geq s_n(P|_\Lambda, \epsilon)$. Hence $\limsup_T \frac{1}{T} \log M_T(\phi|_K, \delta) \geq \frac{1}{\tau_{\max}} \limsup_n \frac{1}{n} \log s_n(P|_\Lambda, \epsilon)$ (as $n/(n-1) \rightarrow 1$). The left side is non-increasing in δ and tends to $h(\phi|_K)$, so $h(\phi|_K) \geq \frac{1}{\tau_{\max}} \limsup_n \frac{1}{n} \log s_n(P|_\Lambda, \epsilon)$ for every $\epsilon > 0$, and letting $\epsilon \rightarrow 0$ gives $h(\phi|_K) \geq h_{\text{top}}(P|_\Lambda)/\tau_{\max}$.

(e) By uniform continuity of $(k, \theta) \mapsto \phi_\theta(k)$ on the compact set $K \times [0, 1]$, for every $\delta > 0$ there is $\delta' > 0$ such that $d(a, b) < \delta'$ implies $d(\phi_\theta a, \phi_\theta b) < \delta$ for all $\theta \in [0, 1]$. If $d(\phi_s x, \phi_s y) \geq \delta$ with $s = k + \theta$, k an integer, $0 \leq k \leq T$, $\theta \in [0, 1]$, then $d(\phi_k x, \phi_k y) \geq \delta'$. So a (T, δ) -separated set of the flow is $(\lfloor T \rfloor + 1, \delta')$ -separated for ϕ_1 , $s_{\lfloor T \rfloor + 1}(\phi_1|_K, \delta') \geq M_T(\phi|_K, \delta)$, and $h_{\text{top}}(\phi_1|_K) \geq h(\phi|_K)$. Finally a separated set of $\phi_1|_K$ is separated for ϕ_1 on $M_E \supset K$. $\square$

Remark 1. With Abramov's formula for a flow under a roof function [1] (stated in [16, Sect. 2]) and the variational principle one obtains the bound with $\tau_{\max}$ replaced by the mean return time of an invariant measure of $P|_\Lambda$ with entropy at least $\log r$ (a lift of the Parry measure through the factor map onto Σ_A ; a measure of maximal entropy is not known to exist, since Λ is not shown to be expansive), which is at least as good; the proofs here do not use it.

5.2 Topological horseshoes and entropy: proof of Corollary 1

Theorem 3 (proved in Sect. 6 from the covering relations, Proposition 1 and Lemma 7) gives, at each of the three energies, the compact invariant set, the semiconjugacy onto a subshift of finite type with $\log r_m > 0$, and the bound for the flow. $\square$

Remark 2 (Smale's theorem and its later forms). Smale's homoclinic theorem [25, Thm. B, p. 64] reads: "If $D(M)$ denotes the space of diffeomorphisms of M with the C^r topology, $r > 0$, there is a subset D_0 which is the countable intersection of open dense sets of D with the following property. If $x \in M$ is a homoclinic point of $T \in D_0$, then there is a Cantor set $\Omega \subset M$, $x \in \Omega$, and p such that $T^p \Omega = \Omega$ and T^p restricted to Ω is equivalent to a shift automorphism of symbolic dynamics." It is stated for a residual set D_0 of diffeomorphisms of a compact manifold, and Smale's proof (Sect. 9, p. 78) first assumes, "without loss of generality", that the periodic points satisfy "Sternberg's non-degeneracy condition", so that "in the neighborhood of a point of M of period p , there will exist local coordinates in which T^p is linear". For an area-preserving map this condition fails at every hyperbolic periodic point ($\lambda_1 \lambda_2 = 1$ gives the resonances $\lambda_1 = \lambda_1^{k+1} \lambda_2^k$, $k \geq 1$), so Smale's own 1965 proof does not apply to our return map as written. Later proofs of the Smale–Birkhoff homoclinic theorem do without the linearization and apply to any C^1 diffeomorphism near a transversal homoclinic orbit, with no non-resonance condition, and locally, so that a map defined only on an open set is no obstacle: for example [14, Thm. 6.5.5], [21, Ch. 2] and, for planar maps, [18]. We have not read these statements here and do not use them. With Theorems 1 and 2 they would give a hyperbolic horseshoe for an iterate of the return map, and positive topological entropy, at the three energies and at every energy of $[-10^{-10}, 10^{-10}]$. The paper's route is the direct one: the covering relations of Theorem 3 give topological horseshoes with explicit entropy bounds at the three energies, from what is written out and computed here.

5.3 No analytic integral on the level: proof of Corollary 2

We first prove the local λ -lemma that the argument needs, from the same cone conditions. Let $\hat{G} = A^{-1} D G A$. Since G is linear and $G p_0 = p_0$ (p_0 lies on $\theta_2 = 0$), $G(p_0 + A\zeta) = p_0 + A\hat{G}\zeta$, and $W_{\text{loc}}^s(p) = G(W_{\text{loc}}^u(p))$ is $\hat{G}(\text{gr } w)$ in local coordinates.

Lemma 8 (a local λ -lemma; proved). *Assume the hypotheses of Lemma 5, with p symmetric and $p_0 \in \text{Fix}(G)$. Let D' be a C^1 arc through a point $s \in W_{\text{loc}}^s(p)$, transversal there to $W_{\text{loc}}^s(p)$. Then for every $\varepsilon > 0$ there is n_0 such that for every $n \geq n_0$ the image $f^n(D')$ contains an arc $\{p_0 + A(x, h_n(x)) : |x| \leq a\}$ with h_n of class C^1 , $\max |h_n - w| < \varepsilon$ and $\max |h'_n - w'| < \varepsilon$.*

Proof. We work in local coordinates, $W = \text{gr } w$, $W^s = \hat{G}(W)$, $s = \hat{G}\zeta^s$ with $\zeta^s \in W$, and $e = a(\mu_h - 1)/(\mu_h + 1)$.

Step 1: the orbit of s . By Lemma 2, $\hat{f}(\hat{G}\zeta') = \hat{G}\hat{f}^{-1}(\zeta')$ whenever $\hat{f}^{-1}(\zeta')$ is defined; so, by Lemma 5(d), $s_n = \hat{f}^n(s) = \hat{G}\hat{f}^{-n}(\zeta^s)$ is defined for all $n \geq 0$, $\hat{f}^n$ is a diffeomorphism near s that maps W^s near s into W^s , and $s_n \rightarrow \hat{G}\zeta_p = \zeta_p$. By (C3) and (C4), $\zeta_p \in \text{int } N_0$ and $|x_p| < e$, so there are a neighbourhood U of ζ_p with $U \subset \text{int } N_0 \cap \{|x| < e\}$ and an n_1 with $s_n \in U$ for all $n \geq n_1$. Replacing D' and s by a sub-arc of $\hat{f}^{n_1}(D')$ and s_{n_1} (still transversal to W^s there), we may assume $s_n \in U$ for all $n \geq 0$.

Step 2: the tangent enters the cone. Let $v \neq 0$ be tangent to D' at s and $v^s = \hat{G}(1, w'(\zeta_x^s))$, tangent to W^s at s . Choose $v^u \in C_u$ not parallel to v^s ; as v is not parallel to v^s , $v = cv^u + c'v^s$ with $c \neq 0$. First, $D\hat{f}^n(s)v^s = \hat{G}D\hat{f}^{-n}(\zeta^s)(1, w'(\zeta_x^s))$ is $\hat{G}$ applied to a tangent vector of W whose x -component has modulus at most μ^{-n} (by Lemma 5(a), φ_w^{-1} has Lipschitz constant μ^{-1}), so $|D\hat{f}^n(s)v^s| \leq C\mu^{-n}$ for a constant C . Second, by (C1) at $s_0, \dots, s_{n-1} \in N_0$, $D\hat{f}^n(s)v^u = (X_n, Y_n)$ with $|X_n| \geq \mu^n |v_x^u|$, $v_x^u \neq 0$, and $|Y_n| \leq \beta |X_n|$ for $n \geq 1$, where $\beta < \alpha$ is the largest ratio $|w|/|u|$ in (C1) over $\zeta \in N_0$ and $|t| \leq \alpha$ (it is attained, by continuity and compactness, and is less than α by (C1)). So $D\hat{f}^n(s)v = c(X_n, Y_n) + O(\mu^{-n})$ has slope of modulus at most $(\beta|c||X_n| + C'\mu^{-n})/(|c||X_n| - C'\mu^{-n}) \rightarrow \beta < \alpha$, and for some n_2 it lies in the interior of C_u . Replacing D' and s by a sub-arc of $\hat{f}^{n_2}(D')$ and s_{n_2} , we may assume that the tangent of D' at s lies in the interior of C_u .

Step 3: growth to full width. By continuity of the tangent, a sub-arc $D_0 \subset D' \cap N_0$ through s is the graph of a C^1 function with slopes in $[-\alpha, \alpha]$ over an interval $J_0 \subset [-a, a]$ with s_x in its interior. Suppose that $D_n \subset \hat{f}^n(D')$ is such a graph over $J_n \subset [-a, a]$, contained in N_0 and through s_n . The argument of Lemma 5(a), with J_n in place of $[-a, a]$, shows that $\hat{f}(D_n)$ is the graph of a C^1 function with slopes in $(-\alpha, \alpha)$ over an interval $\varphi(J_n)$ that contains $(s_{n+1})_x$, and that each end of $\varphi(J_n)$ is the image of an end of J_n and at least μ_h times as far from $(s_{n+1})_x$ as that end is from $(s_n)_x$. Let D_{n+1} be its part over $J_{n+1} = \varphi(J_n) \cap [-a, a]$; $D_{n+1} \subset N_0$ by (C2), and $s_{n+1} \in D_{n+1}$. If an end of J_n is an end of $[-a, a]$, it is at distance at least $a - |(s_n)_x| > a - e$ from $(s_n)_x$, so its image is at distance greater than $\mu_h(a - e) = a + e > a + |(s_{n+1})_x|$ from $(s_{n+1})_x$, and lies outside $[-a, a]$. So an end of J_{n+1} that lies inside $(-a, a)$ is the image of an end of J_n inside $(-a, a)$, and its distance from $(s_{n+1})_x$ is at least μ_h times that of the latter from $(s_n)_x$. These distances are at most $2a$ and $\mu_h > 1$, so after finitely many steps $J_n = [-a, a]$, and from then on $D_n = \text{gr } h_n$ with $h_n \in \mathcal{L}$ of class C^1 and $h_{n+1} = \mathcal{H}h_n$. By Lemma 5(c), with the fixed point w of (d), $h_n \rightarrow w$ and $h'_n \rightarrow w'$ uniformly. $\square$

This is the case of Palis's λ -lemma [20, Lemma 1.1, p. 387] that the argument needs. Palis states it for diffeomorphisms of a compact manifold; f is defined only on an open subset of the plane, so we prove the local case from the certified cone conditions instead of applying his statement.

The proof is Kozlov's argument [15, Ch. V, §2, proof of Thm. 3], written out for our setting. Let F be real-analytic near M_E and invariant under the flow, and let $g = F|_{\Sigma_E}$, a real-analytic function of (θ_2, p_2) on A_E (Lemma 1) with $g \circ P = g$ where P is defined; as g is 2π -periodic in θ_2 , also $g \circ f = g$ in the lift. Let $c = g(p)$. If $z' \in W^u(p)$, then $g(z') = g(P^{-n}z') \rightarrow g(p)$, so $g = c$ on $W^u(p)$, and likewise on $W^s(p)$.

Let $z \in W_{\text{loc}}^u(p)$ be the point of Lemma 6 with $f^k(z) = q$. As $Gq = q$ in the lift ($m = 0$, Sect. 4.4), Lemma 2 gives $f^k(q) = Gf^{-k}(q) = Gz$, so f^{2k} is defined at z and, its domain being

open, on a short sub-arc $\Gamma' \subset W_{\text{loc}}^u(p)$ around z . Let $D = f^k(\Gamma')$, a compact C^1 arc in $W^u(p)$ through q that does not contain p . By Lemma 6, D is transversal at q to the arc $G(D) \subset W^s(p)$, and by Lemma 2 f^k maps $G(D)$ onto $G(f^{-k}(D)) = G(\Gamma') \subset W_{\text{loc}}^s(p)$. So $D' = f^k(D)$ is a C^1 arc through $f^k(q) = Gz \in W_{\text{loc}}^s(p)$, transversal there to $W_{\text{loc}}^s(p)$. In the local coordinates of Lemma 5, fix $x^b \neq x_p$ in $(-a, a)$ and, for x in a small interval I around x^b not containing x_p , the vertical segments $\sigma_x = \{(x, y) : |y - w(x)| \leq \epsilon\}$, which are real-analytic arcs crossing the graph of w transversally at $(x, w(x))$. By Lemma 8, applied to D' , for all large n the image $f^{k+n}(D)$ contains an arc D_n , the graph of h_n over $[-a, a]$, and $h_n \rightarrow w$ uniformly. So D_n meets σ_x at a point $z_n(x) \rightarrow (x, w(x))$. Moreover $z_n(x) \neq (x, w(x))$ for large n : otherwise $f^{-(k+n)}(x, w(x)) \in D$ for infinitely many n , while $f^{-(k+n)}(x, w(x)) \rightarrow p$ by Lemma 5(d), and p is not in the compact arc D . Hence the $z_n(x)$ take infinitely many values, all different from $(x, w(x))$ for large n , and they converge to it. Since $z_n(x) \in W^u(p)$, $g(z_n(x)) = c$. So the real-analytic function $y \mapsto g(x, y) - c$ on the segment σ_x has zeros accumulating at $w(x)$ and vanishes identically. Hence $g = c$ on the open set $\bigcup_{x \in I} \sigma_x$, $F = c$ on its flow-box, which is open in M_E , and $F = c$ on M_E because M_E is a connected real-analytic manifold (Lemma 1). $\square$

5.4 No global analytic integral: proof of Corollary 3

H is proper and $M_0 \subset U$ is compact, so there is $\delta > 0$ with $M_E \subset U$ for $|E| < \delta$. Let $I = (-\min(\delta, 10^{-10}), \min(\delta, 10^{-10}))$. By Theorem 2 and Corollary 2, F is constant on each M_E , $E \in I$. So on the open set $H^{-1}(I)$ the differential of F vanishes on the tangent spaces of the level sets, $dF = \varphi dH$ there (with $dH \neq 0$ by Lemma 1), and $dF \wedge dH = 0$ on $H^{-1}(I)$. The 2-form $dF \wedge dH$ is real-analytic on the connected set U and vanishes on a non-empty open subset, hence everywhere. $\square$

Remark 3. Theorem 2 replaces the standard persistence argument (a transversal homoclinic point of an analytic family persists for nearby parameters) by a computation, so that Corollary 3 rests only on what is verified.

6 Explicit topological horseshoes

6.1 Covering relations

We use the covering relations of Zgliczyński and Gidea [33] with one expanding direction. An h-set in $\mathbb{R}^2$ here is a parallelogram $X = c_X + B_X[-1, 1]^2$ with an invertible matrix B_X whose first column spans the nominally expanding direction; its normalized coordinates are $(x, y) = B_X^{-1}(z - c_X)$, its left and right edges are $X^{le} = \{x = -1\}$ and $X^{re} = \{x = 1\}$, $X^+ = \{|x| \leq 1, |y| = 1\}$, and $S(X)^l = \{x < -1\}$, $S(X)^r = \{x > 1\}$. For a continuous map h on X , [33, Thm. 16] states: if for some $|q_0| \leq 1$

- (i) $h(\{(x, q_0) : |x| \leq 1\}) \subset \text{int}(S(Y)^l \cup Y \cup S(Y)^r)$ (their (76)),
- (ii) $h(X) \cap Y^+ = \emptyset$ (their (77)), and
- (iii) either $h(X^{le}) \subset S(Y)^l$ and $h(X^{re}) \subset S(Y)^r$, or $h(X^{le}) \subset S(Y)^r$ and $h(X^{re}) \subset S(Y)^l$ (their (78), (79)),

then X h -covers Y with degree ± 1 (their Definition 6), written $X \xrightarrow{h} Y$. Their Theorem 9 states that for a closed chain $X_0 \xrightarrow{h_1} X_1 \Rightarrow \cdots \xrightarrow{h_k} X_k = X_0$ of such relations there is $x \in \text{int } X_0$ with $h_i \circ \cdots \circ h_1(x) \in \text{int } X_i$ for $1 \leq i \leq k$ and $h_k \circ \cdots \circ h_1(x) = x$.

6.2 From covering relations to entropy

Proposition 1 (proved). *Let $X_1, \dots, X_n$ be h -sets in the lift whose projections to the cylinder are pairwise disjoint, and let $\mathcal{G}$ be a strongly connected directed graph on $\{1, \dots, n\}$ with adjacency matrix A , such that for every edge $i \rightarrow j$ there is an integer k_{ij} with $X_i \xrightarrow{h_{ij}} X_j$ for $h_{ij} = T^{k_{ij}} \circ P$, where P is continuous and injective on a neighbourhood of each X_i . Let Λ be the set of points of $\bigcup_i \pi(X_i)$, with π the projection to the cylinder, that have a full P -orbit in $\bigcup_i \pi(X_i)$ visiting the sets along edges of $\mathcal{G}$. Then Λ is compact and P -invariant, $P|_\Lambda$ is a homeomorphism, the itinerary map $\iota : \Lambda \rightarrow \Sigma_A$ is a continuous surjection with $\iota \circ P = s \circ \iota$ (s the shift), and if $Av \geq rv$ for some $r > 0$ and a vector $v > 0$, then $h_{\text{top}}(P|_\Lambda) \geq \log r$.*

Proof. Write X_i also for $\pi(X_i)$ and let $\delta > 0$ be the smallest distance between two of these compact, pairwise disjoint sets. *Periodic orbits.* For a closed path $i_0 \rightarrow i_1 \rightarrow \dots \rightarrow i_k = i_0$ in $\mathcal{G}$, Theorem 9 of [33] applied to the relations $X_{i_{j-1}} \Rightarrow X_{i_j}$ gives a point whose h -orbit, hence (projected) whose P -orbit, is periodic and visits the sets in this order. *Every bi-infinite path is realized.* Let $(s_j)_{j \in \mathbb{Z}}$ be a path. Since $\mathcal{G}$ is strongly connected, for each n the segment $s_{-n}, \dots, s_n$ is part of a closed path, which gives a periodic orbit and points $z_j^{(n)} \in X_{s_j}$, $|j| \leq n$, with $P(z_j^{(n)}) = z_{j+1}^{(n)}$. By compactness and a diagonal argument, a subsequence has $z_j^{(n)} \rightarrow z_j \in X_{s_j}$ for every j , and by continuity $P(z_j) = z_{j+1}$. So $z_0 \in \Lambda$ with itinerary (s_j) . *Compactness and continuity.* Full orbits in $\bigcup X_i$ are unique given their point at time 0, by injectivity. If $z^{(n)} \in \Lambda$ converge to z , the same diagonal argument applied to their orbits (whose itineraries converge coordinatewise along a subsequence, the symbols being finitely many) gives a full orbit of z along edges of $\mathcal{G}$; so Λ is closed, hence compact. $P(\Lambda) = \Lambda$, and $P|_\Lambda$ is a continuous bijection of a compact space, a homeomorphism. ι is continuous because each $X_i \cap \Lambda$ is open and closed in Λ , and it is onto by the previous step. *Entropy.* Since $A \geq 0$, $A^{n-1}v \geq r^{n-1}v$, so the number of paths with n vertices is $\mathbf{1}^\top A^{n-1} \mathbf{1} \geq \mathbf{1}^\top A^{n-1}v / \max v \geq r^{n-1} \mathbf{1}^\top v / \max v \geq r^{n-1}$. For each such path choose a point of Λ whose itinerary begins with it. Two of these points with different paths are, at some time $j < n$, in different sets, at distance at least δ : they are (n, δ) -separated. By Bowen's definition of topological entropy through separated sets [29, Sect. 7.2], $h_{\text{top}}(P|_\Lambda) \geq \limsup_n \frac{1}{n} \log r^{n-1} = \log r$. $\square$

6.3 The computation and the proof of Theorem 3

The h -sets were designed numerically by `horseshoe_design.cpp`. The configurations are `horseshoe_E0.cfg`, `horseshoe_Ehalf.cfg` and `horseshoe_Eminushalf.cfg`. At $E = 0$: a pseudo-orbit along the homoclinic loop, built forward from the point $P^{-9}(q_0)$ of the local unstable manifold for nine returns and completed by the reversibility, $z_{18-i} = Gz_i$, then continued along the stable manifold towards p_0 for three returns; expanding directions pushed forward by DP and contracting directions $s_i = DG u_{18-i}$; widths chosen so that each image overshoots the next set by a factor of about 3, and thicknesses three times the sampled image thickness but not below 10^{-8} . None of this design is part of the proof. The set N is a parallelogram of half-sizes 10^{-5} along v_u and $2.5 \cdot 10^{-7}$ along v_s around p_0 . At $E = \pm 1/2$ the same construction starts from $P^{-11}(q_E)$ (eleven returns, the k of Table 1), which gives 27 sets; N has half-sizes $5.5 \cdot 10^{-6}$ and $2.6 \cdot 10^{-7}$ at $E = 1/2$, and $7.5 \cdot 10^{-6}$ and $4 \cdot 10^{-7}$ at $E = -1/2$. At $E = -1/2$, where $|\lambda_u| = 2.885$ is below 3, the factor-3 margins would make the widths grow backwards and the thicknesses grow forwards along the chain, so the design uses an overshoot factor 2.8 and a thickness factor 2. The design commands are in `code/designs.sh`, which regenerates the three configurations into a temporary folder and compares them with the committed files (byte for byte on the machine used here); `code/run_all.sh` checks the committed files, so a different floating-point platform cannot

change what is verified. A first design at $E = 1/2$ with thickness $1.3 \cdot 10^{-7}$ for N failed the last relation, and a first design at $E = -1/2$ with the factor-3 margins failed the last relation and the disjointness of M_0 and M_{25} ; they are not used.

The program `horseshoe_check.cpp` verifies, for each relation $N \Rightarrow N$, $N \Rightarrow M_0$, $M_i \Rightarrow M_{i+1}$ ($0 \leq i \leq m-2$) and $M_{m-1} \Rightarrow N$ (24 relations at $E = 0$, 28 at $E = \pm 1/2$), with the map $T^k \circ P$ for the integer k in the configuration: condition (i) with $q_0 = 0$, in the stronger form that the image of the midline has $|y| < 1$ in the normalized coordinates of the target; condition (ii) on a cover of the whole set; and condition (iii). Each covering uses 16 pieces per edge and per midline and 64 pieces for the set, bisected up to depth 4 where the check does not resolve, and the mean-value form of Sect. 4.4 with CAPD's C^0 enclosure at a centre point and C^1 enclosure over the piece; every piece must have an enclosed first return, which also makes P defined and continuous on every set. The program also verifies that the sets are pairwise disjoint on the cylinder, bounds the return time over all of them, builds the graph from the relations it has verified (not from the configuration), and checks $Av \geq r_c v$ for a numerical Perron vector $v > 0$, with r_c the lower bound of $\min_i (Av)_i / v_i$ evaluated in interval arithmetic. Bounds are printed rounded in the safe direction. The reports are `data/horseshoe_E0.txt`, `data/horseshoe_Ehalf.txt` and `data/horseshoe_Eminushalf.txt`: at each energy every relation covers, the sets are pairwise disjoint, and all the checks pass (at $E = 0$ the largest value of $|y|$ on a midline image is 0.503, for $M_{21} \Rightarrow N$). The return-time bounds printed are 7.355385354, 6.970677596 and 4.249825826; the runs took about 12, 12 and 13 minutes on two threads.

Proof of Theorem 3. At each energy the verified relations are covering relations by [33, Thm. 16]; the graph is strongly connected (every vertex lies on the loop $N \rightarrow M_0 \rightarrow \dots \rightarrow M_{m-1} \rightarrow N$), and the sets are pairwise disjoint on the cylinder. Let $r = r_m > 1$ be the positive root of $r^{m+1} = r^m + 1$ (there is exactly one, by Descartes' rule of signs): $r_{22} = 1.10695024501688\dots$, $\log r_{22} = 0.1016087069\dots$, and $r_{26} = 1.09483270790531\dots$, $\log r_{26} = 0.0906015734\dots$. The vector v with $v_N = 1$ and $v_{M_i} = r^{-(m-i)}$ ($0 \leq i \leq m-1$) is positive and satisfies $Av = rv$ exactly: $(Av)_{M_i} = v_{M_{i+1}} = rv_{M_i}$ for $i \leq m-2$, $(Av)_{M_{m-1}} = v_N = rv_{M_{m-1}}$, and $(Av)_N = v_N + v_{M_0} = 1 + r^{-m} = r$. Proposition 1 gives Λ_E , the semiconjugacy and $h_{\text{top}}(P_E|_{\Lambda_E}) \geq \log r_m$. (The program's bounds r_c , computed from the relations it verified, agree: $r_c \geq 1.106950245015$ at $E = 0$ and $r_c \geq 1.094832707903$ at $E = \pm 1/2$.) For the flow: Λ_E is compact, $P_E(\Lambda_E) = \Lambda_E$ and $P_E|_{\Lambda_E}$ is continuous and injective (Proposition 1); τ is continuous on the sets, on which every piece has an enclosed transversal first return; and $\max_{\Lambda_E} \tau \leq \tau_E$, because the program's return-time bound is the largest upper end of CAPD's return-time enclosures over pieces that cover every one of the sets (each set is the source of a verified relation). Lemma 7 gives $h_{\text{top}}(\phi_1) \geq \log r_m / \tau_E$, which is > 0.0138141 , > 0.0129975 and > 0.0213188 at $E = 0, 1/2$ and $-1/2$. $\square$

7 Controls and cross-checks

Negative controls (computer-assisted runs that must fail, and do). `run_all.sh` runs nine controls; their reports are in `data/`. The first four are coarse: they show that a check is evaluated and can fail, not that it fails for the reason it is meant to test. The five near-miss controls perturb one quantity just past what the corresponding condition allows, and `run_all.sh` checks in each report that the failure is the intended one and, where stated, the only one.

- *The integrable limit* (coarse, but meaningful for (C6)). Two uncoupled pendulums, $H_0 = p_1^2/4 + p_2^2/2 - 2\cos\theta_1 - \cos\theta_2$, at $E = 0$ (`control_uncoupled_E0`). Stages 1 and 2 pass: the upright lower pendulum, $\theta_2 = \pi$, $p_2 = 0$, is a hyperbolic fixed point of the return map

(trace about 848.1). Stage 3 fails, as it must: the unstable manifold is the separatrix $p_2 = 2|\cos(\theta_2/2)|$, which meets $\text{Fix}(G)$ at $\theta_2 = 2\pi$ with a horizontal tangent, the -1 eigendirection of DG , where it coincides with the stable manifold; the certified slope interval still contains 0 after 16 bisections. The method does not certify a transversal crossing in an integrable system.

- *Mutations of the configuration* (coarse). Moving the segment $[x_1, x_2]$ so that both edge images lie on the same side of the line fails (C5) (`control_mut_segment`); narrowing the cone to $\alpha = 10^{-4}$, below the largest ratio $|w|/|u| \approx 2.92 \cdot 10^{-4}$ measured with that cone, fails (C1) (`control_mut_alpha`).
- *Covering relations that must not hold* (coarse). A skipped step $M_0 \Rightarrow M_2$, a backward step $M_2 \Rightarrow M_1$ and $M_5 \Rightarrow M_5$ all fail (`control_horseshoe_wrong`); their images land far from their targets, so they would fail whatever the details of conditions (i)–(iii).
- *Near-miss covering relations*, from the true relation $M_1 \Rightarrow M_2$ at $E = 0$. With the contracting column of the target shrunk tenfold, below the thickness of the image, only condition (ii) fails: the edges are still mapped across and the midline stays inside, but 24 pieces of the image meet Y^+ (`control_hs_thin`). With the expanding column of the target enlarged tenfold, beyond the stretch of the image, only condition (iii) fails (`control_hs_wide`). With a third h-set that overlaps M_1 (moved by half its expanding column), the relation still holds and only the disjointness test fails (`control_hs_overlap`).
- *Near misses of (C4)*. With the columns of A exchanged and N_0 a square of half-side b , so that graphs are taken over the stable direction on a box small enough for $[M]_{11}$ to exclude 0, (C4) fails for the real reason: $\mu_h \geq 0.2831$, about $|\lambda_s|$, and $\kappa \leq 3.54$, vertical expansion instead of contraction (`control_cones_swap`). With N_0 centred at $p_0 + 0.6a v_u$, so that $|x_p| \approx 0.6a = 1.5 \cdot 10^{-5}$, only the last inequality of (C4) fails, against $a(\mu_h - 1)/(\mu_h + 1) \geq 1.394 \cdot 10^{-5}$ (`control_cones_shift`).

An independent cross-check of the fixed point (computer-assisted). The folder `code/crosscheck` contains a rigorous Taylor integrator in ball arithmetic (Arb, through python-flint), with Hamilton’s equations derived again from (2) by SymPy, validated step enclosures with Lagrange remainders, rigorous detection of the upward crossings of $\theta_1 = 0$ and the variational equation, written without CAPD. At $E = 0$ it verifies the Krawczyk condition on a box of radius 10^{-17} (`kraw3.py`): the fixed point is on $\text{Fix}(G)$, lies inside the enclosure of Theorem 1, has trace $-3.808656368 \pm 8.3 \cdot 10^{-10}$, determinant $1 \pm 1.2 \cdot 10^{-9}$ and $\lambda_u = -3.52496566 \pm 2.5 \cdot 10^{-9}$. At $E = \pm 1/2$ it encloses, at the configured points, the return time, trace and multiplier (`otherE.py`), in agreement with Theorem 1; these are point evaluations, not a second proof.

Numerical cross-check of the crossing (numerical). With the same integrator used at high precision without error control (`edges_nr.py`), the images after 9 returns of the two edge points at $E = 0$ have $\theta_2 = 1.52049688 \cdot 10^{-3}$ and $-3.11657765 \cdot 10^{-3}$ (inside the certified intervals of Table 1), the crossing is at $x = -1.88595114 \cdot 10^{-5}$ with $p_2 = 0.822148135$, and the slope of $W^u(p_0)$ there is $dp_2/d\theta_2 = -1.336202144$, inside the certified $[-6.2138, -0.5237]$.

How the orbit was found (numerical). `scan.cpp` finds symmetric periodic orbits by shooting along $\text{Fix}(G)$; the orbit used is, at each energy, the symmetric fixed point of P with the smallest multiplier found. `manifold.cpp` samples a fundamental domain of $W^u(p)$, iterates it and reports

its crossings of $\text{Fix}(G)$ with their slopes; at $E = 0$ the first crossing appears after 11 returns from a distance 10^{-6} , and the proof uses the same crossing two returns later on a fundamental domain farther out. Poincaré sections (`explore.cpp`) of 60 random orbits of 400 returns show a visible island at $E = -1/2$ and none at plot resolution at $E = 0$ and $E = 1/2$ (an exploration whose plots are not recorded).

8 Discussion

What is proved. At the three energies and on an interval of energies around 0, the classical double pendulum with equal masses and lengths has a hyperbolic periodic orbit with a transversal homoclinic orbit; hence no non-constant analytic invariant function on these levels, and no real-analytic first integral independent of the energy on any connected open set containing M_0 . At $E = -1/2$, 0 and $1/2$ it has topological horseshoes and positive topological entropy, with explicit bounds from covering relations; on the interval of energies no horseshoe is proved here (Remark 2). The bounds are lower bounds from a single loop and are presumably far below the entropy of the flow, which we have not estimated numerically; nothing is claimed about the measure of the chaotic set, about energies other than those stated, or about ergodicity.

Relation to Bolotin and Negrini. Bolotin and Negrini [2] prove a variational criterion for non-integrability of analytic Lagrangian systems with two degrees of freedom on a torus or a cylinder, and apply it in their Section 10 to the double pendulum with arbitrary masses and lengths, near the energy level $h = \max V$ of the upright equilibrium ($E = 3$ in our units, far above our energies). We know this paper only from search snippets: we have not seen the printed paper, and read parts of its Sections 3 and 10 in the search-within snippets of the journal volume that are openly available. The following rests on that text, and the printed paper has to be read before the novelty stated here can be relied on. Their Theorem 10.1 is derived from the condition $2\pi\mu < 2d$ (p. 435), with $\mu^2 = 2g(m_1l_1^2 + m_2(l_1 - l_2)^2)((m_1 + m_2)l_1 + m_2l_2)$ (p. 434) and $2d = \frac{16}{3}m_2\sqrt{g}(\max\{l_1, l_2\})^{3/2}$ (p. 435); at $m_1 = m_2 = m$, $l_1 = l_2 = l$ it reads $2\pi\sqrt{6} < 16/3$ in units of $m\sqrt{g}l^{3/2}$, that is $15.39 < 5.33$, false by a factor of about 2.9. Squared, the condition is $9\pi^2(m_1l_1^2 + m_2(l_1 - l_2)^2)((m_1 + m_2)l_1 + m_2l_2) < 32m_2^2(\max\{l_1, l_2\})^3$, which at equal lengths requires $m_1/m_2 < 0.166$. Their Section 3 also asserts, in the setting of their criterion, that the double pendulum of their Section 10 has positive topological entropy (p. 420). So, on the reading available to us, neither their non-integrability nor their chaos covers the equal case, although their abstract criterion might, with sharper estimates, which they mention (“We can perform the estimates more carefully and obtain a better condition”, p. 435). The statement of Theorem 10.1 as rendered in the snippets begins with “9m2” where the derivation requires $9\pi^2$; if the printed statement differs from its derivation, only the printed page can tell, and our claim that global analytic non-integrability (Corollary 3) and chaos in the equal double pendulum had not been proved is conditional on this reading. The novelty of the results on the levels $-1/2 \leq E \leq 1/2$ (Theorems 1–3, Corollaries 1 and 2) does not depend on it, since their results concern energies near $E = 3$.

Relation to Ivanov’s conjecture. Ivanov [9, p. 105] conjectured that the double pendulum is non-integrable on every energy level for all non-degenerate masses, lengths and energies, and planned to cover the parameter space by numerically verified neighbourhoods together with asymptotic results near its boundary [9, 11, 12]. At equal masses and lengths, Corollaries 2 and 3 give the real-analytic form of that statement on the levels treated here, with every computation in interval arithmetic.

What is not proved. Meromorphic (Liouville) non-integrability in the sense of Morales-Ruiz and Ramis [19] is not proved here: it concerns complex-analytic integrals on a neighbourhood of a complex phase curve and needs a particular solution along which the differential Galois group of the variational equation has a non-abelian identity component, which, as Stachowiak and Szumiński [27] and Szumiński and Kapitaniak [28] explain, is the obstacle for the classical double pendulum; Salnikov’s numerical monodromy computation [23] has not, as far as we know, been made rigorous. Our Corollary 3 concerns real-analytic integrals and is of a different nature. Only the stated energies are covered. The symmetric periodic orbit was followed numerically between $E = -1/2$ and $1/2$ (an exploration not recorded in `data/`), but the symmetric homoclinic point is not the same one at all three energies: at $E = 0$ and $1/2$ the crossing of Table 1 has $p_2 \approx 0.822$ and 0.824 , while at $E = -1/2$ it has $p_2 \approx -1.466$ and the edge images come in the opposite order, so at $E = -1/2$ the proof uses a different intersection of the same half of $W^u(p)$ with $\text{Fix}(G)$, not the continuation of the points used at the other two energies. A proof on the whole band would need a cover of the energy interval by small intervals, each verified as in Sect. 4.6, and a choice of homoclinic point along it.

What the proof trusts. The proof trusts CAPD’s rigorous integrator and Poincaré map, the compiler and the floating-point rounding of the machine. Our sets start on the section, where CAPD’s documentation says only that the initial point “does not need to be on Poincare section”. That no upward crossing is skipped follows from three facts in the source at the pinned commit. In `integrateUntilSectionCrossing (PoincareMap_templateMembers.h)`, the set is first stepped until the section function θ_1 is negative on the whole of its enclosure at a step end (the departure loop), then until it is not, and the crossing is computed in that step. (a) After every step, `checkTransversability` evaluates θ_1 on the step enclosure (an enclosure of all trajectories over the step) and, if the result contains 0, requires $\dot{\theta}_1$ to be single-signed on it, and fails otherwise. (b) The step enclosure contains the box hull of the set at the start of the step (`HighOrderEnclosure`: a Taylor polynomial in $t \in [0, h]$ whose constant term is that hull, plus a remainder term that contains 0). (c) The sign tested at a step end is θ_1 on the box hull of the set (`CoordinateSection::evalAt`). Suppose a point makes an upward crossing at a time s in step k of the departure loop. It starts on the section with $\dot{\theta}_1 > 0$, so it crossed downward at some $s' < s$, in step $i \leq k$. If $i = k$, $\dot{\theta}_1$ takes both signs on the step enclosure, which meets the section, against (a). Otherwise, at each step end t_j , $i \leq j < k$, the loop went on, so by (c) the range of θ_1 on the hull has a non-negative upper end; it also contains the point’s $\theta_1(t_j) \leq 0$, so it contains 0; by (b) so does θ_1 on the next step enclosure, and by (a) $\dot{\theta}_1$ is single-signed on it. Consecutive step enclosures both contain the point’s state at their common step end, so the signs agree, and from $\dot{\theta}_1 < 0$ on step i it follows that $\dot{\theta}_1 < 0$ on step k , which contains the upward crossing: a contradiction. After the departure loop, a point that crosses upward in a step has $\dot{\theta}_1 > 0$ on it by (a), so $\theta_1 > 0$ at its end, and the loop stops there. So the crossing CAPD returns is, for every point of the set, the first upward crossing after time 0. This rests on our reading of the three facts in the code, not on a documented guarantee; the fixed point at $E = 0$ is also confirmed by the independent Arb computation of Sect. 7.

Data availability. The programs, their configurations and their output are available in the companion repository, <https://github.com/ChaseHendrick/double-pendulum>; release 1.0.2 archives these exact proof files and supplementary checks at <https://doi.org/10.5281/zenodo.23047057>. They accompany this manuscript in the folders `code/`, `configs/` and `data/`: `code/run_all.sh` builds CAPD at the pinned commit and the programs, and reruns the field check, Theorems 1–3 on the committed configurations and the controls (about 85 minutes on two threads of a shared machine, CAPD already built);

`code/designs.sh` regenerates the h-set designs and compares them with the committed configurations; `code/crosscheck` holds the independent Arb computation.

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